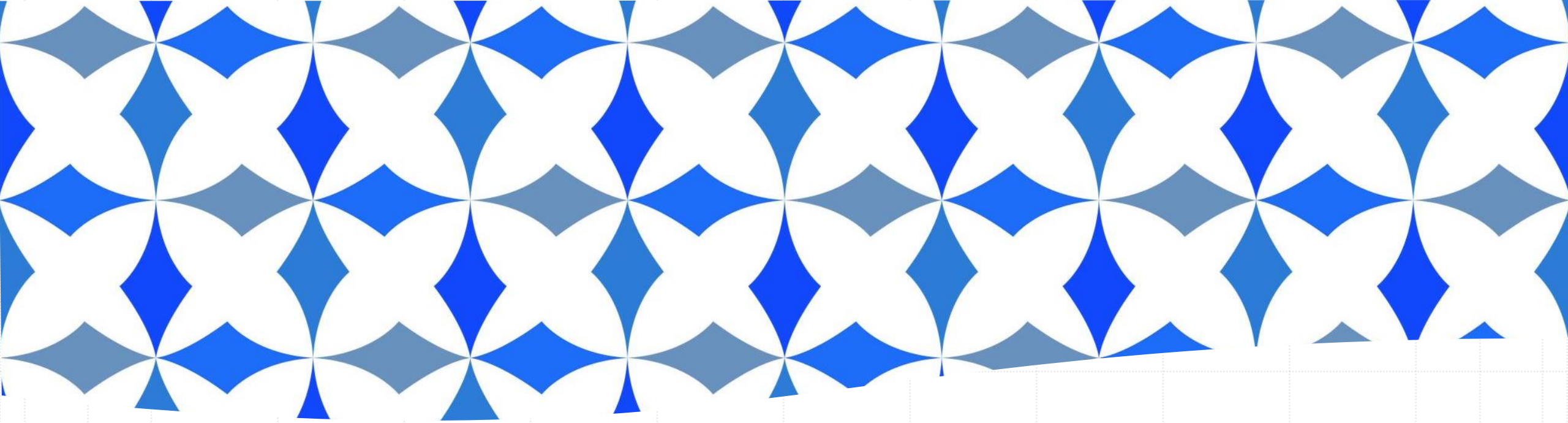




Risk Analysis

ENGCV370

Dr Megan Boston



Defining
Risk

Types of
Risk

Safety in
Design

Quantifying
Risk

Risk

Safety in Design

- Easiest to manage risk earlier in a project
- Managing risk helps to improve safety
- Managing risk improves overall project outcomes

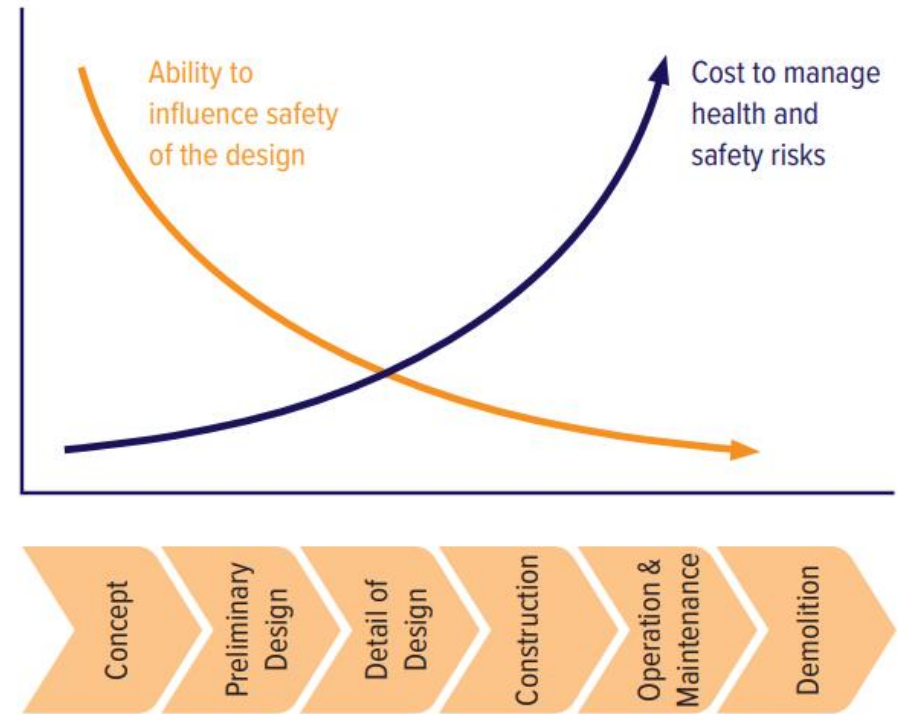
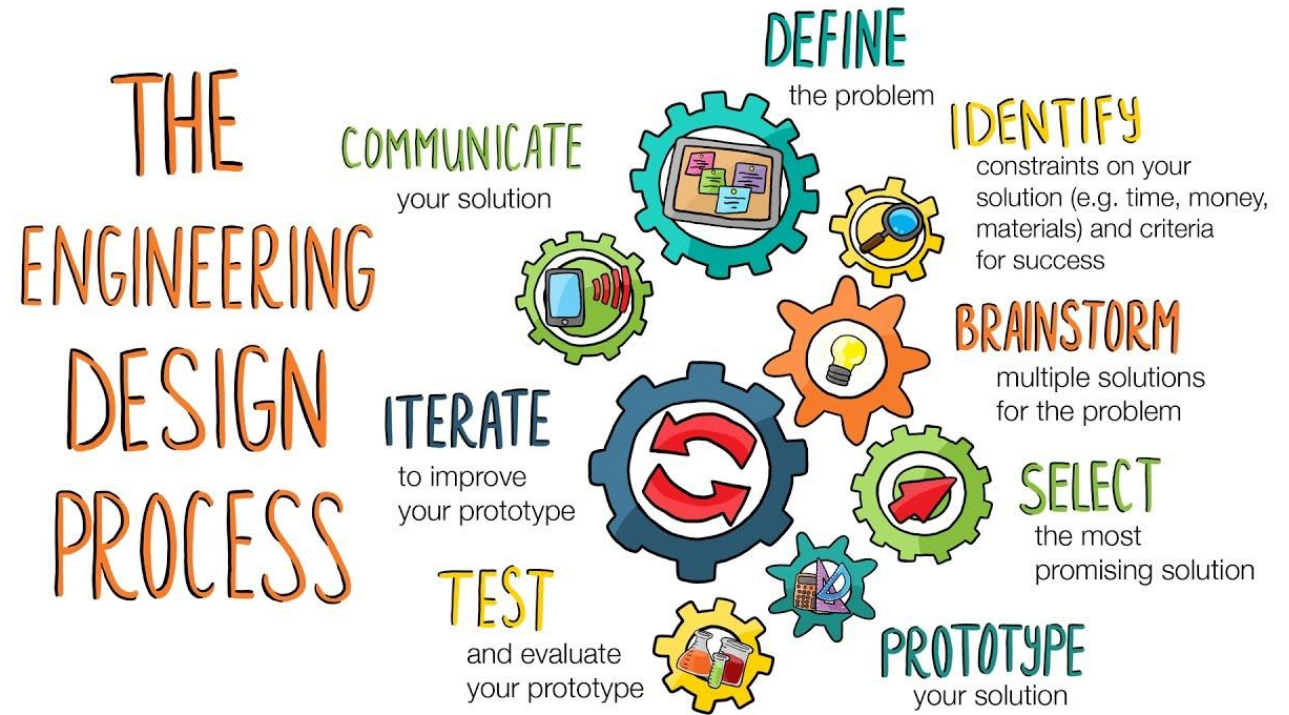


Figure 1: Symberszki chart of influence over a product's life cycle (adapted from Symberszki, R, (1997), Construction Project Safety Planning. TAPPI Journal, 80 (11), 69–74)

Safety in Design

- Elimination of potential hazards at the beginning of the project
- More efficient and effective risk management





Defining Risks

Understanding the final use of the product

- What will the product be used for?
- Who will use it?

Engage with the client and end users

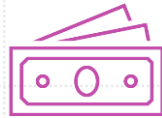
- How will it be used?
- Where will it be located?
- What are all the possible things that can go wrong?
 - Extreme events
 - Failures

Types of Risk



Environmental Risk

Human Health
Ecological
Global Climate Change



Business Risk

Financial
Schedule
Strategic



Technical Risk

Reliability
Catastrophic Failures



Political Risk

Public Acceptance
Public Perception



Classify Risk

Being the lead contractor on a \$600 million-dollar industrial construction project with a fixed-price contract and stiff penalties for completion

Deciding how to design a 47-storey building in an earthquake zone

Deciding where to locate a hazardous waste facility

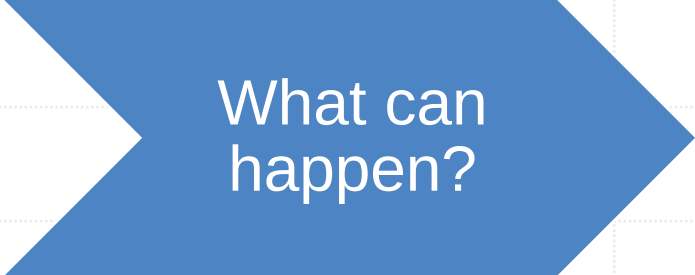
Deciding how to allocated money to protect New Zealand infrastructure

Being the lead project manager of a multi-million-dollar satellite project with a short launch window

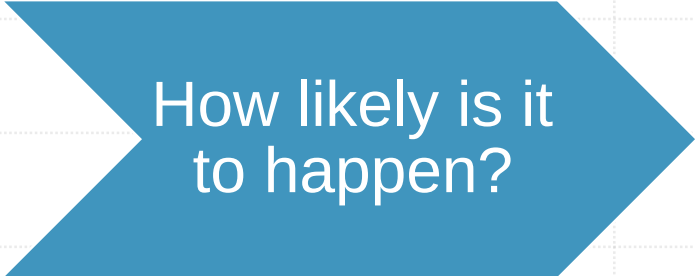


Quantify the risks

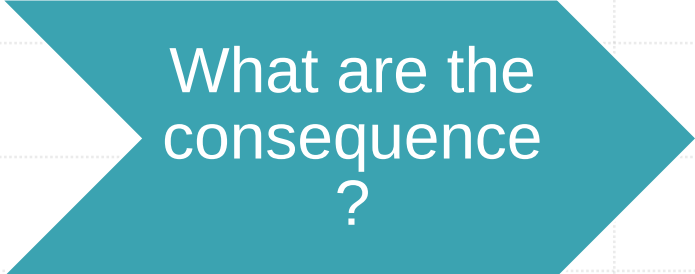
- Scenarios
 - What can go wrong?
- Likelihood (qualitative or quantitative) of the scenario happening
- Consequence (qualitative or quantitative severity of performance degradation) if the scenario happens



What can happen?



How likely is it to happen?



What are the consequence ?

How to Quantify Risk?



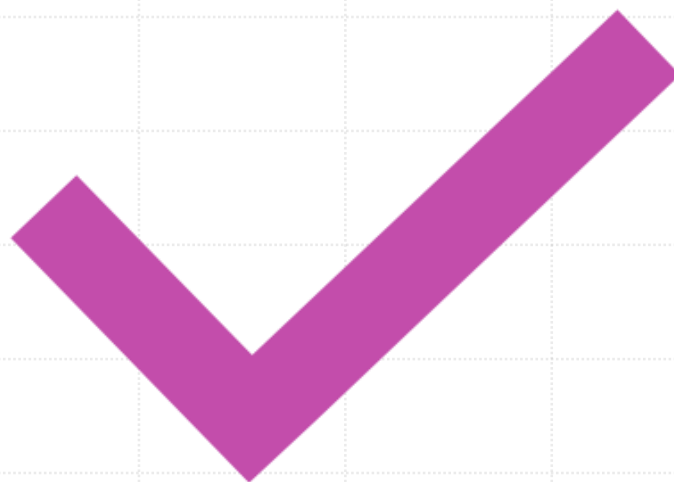
Assessing Risk: Probabilistic Risk Assessment

- Helps to determine risk when there is not a lot of data available
- Steps:
 1. Functional Analysis
 2. Effects of External Events
 3. Probability Calculations
 4. Use Results to Make Decisions



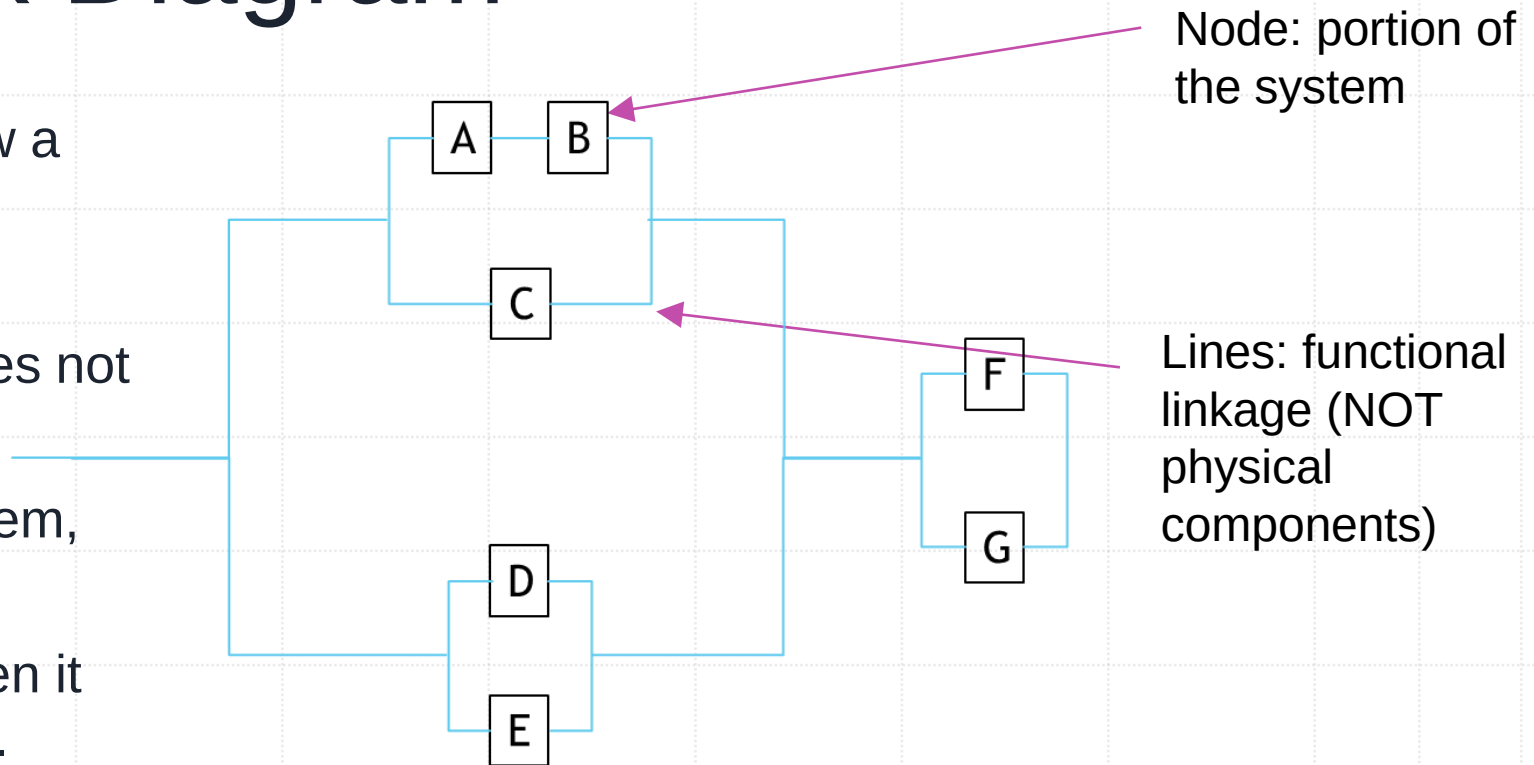


Functional Analysis



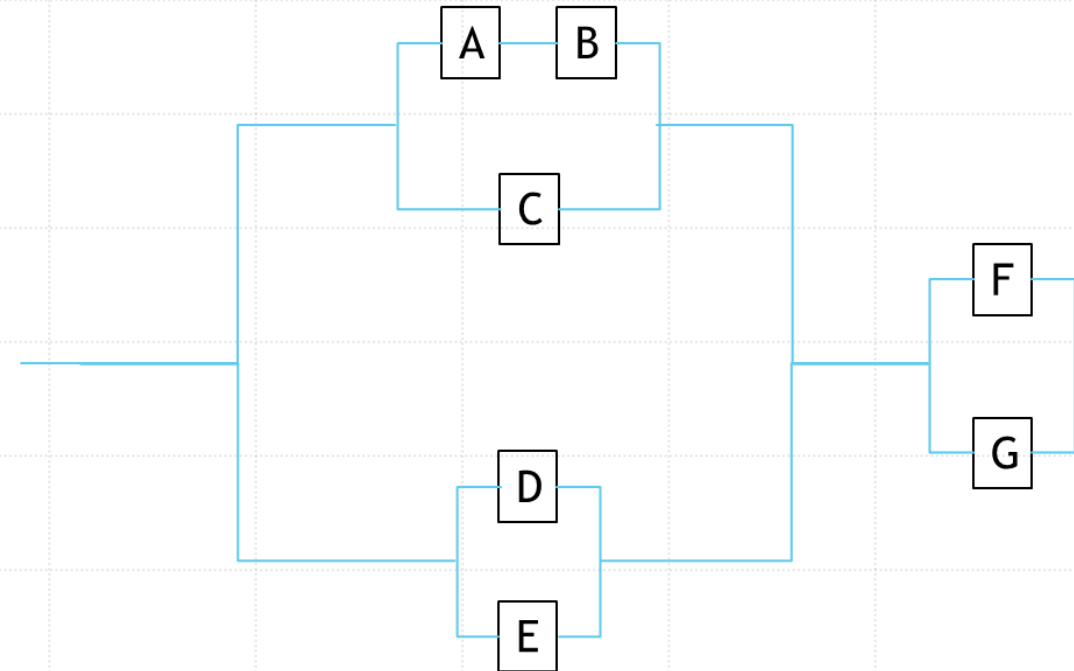
Functional Block Diagram

- Method to graphically display how a system functions
 - System logic diagram
- Binary: the system works or it does not work
- If there is a path through the system, then the system will function
- If a portion of the system fails, then it blocks the path through that area.



Functional Block Diagram

- Tips for Creating a Functional Block Diagram
 1. If two or more components need to work they are in series
 2. If only one of two or more components need to work they are in parallel

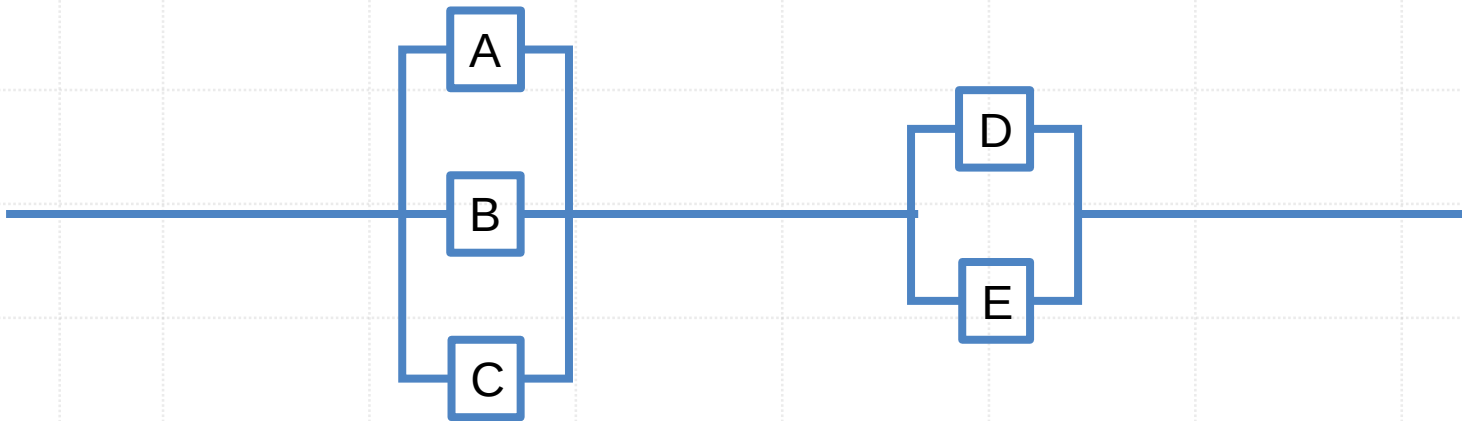




Create functional block diagrams

- A system works if one of A, B and C works AND either D or E works

A system works if one of A, B and C works
AND either D or E works



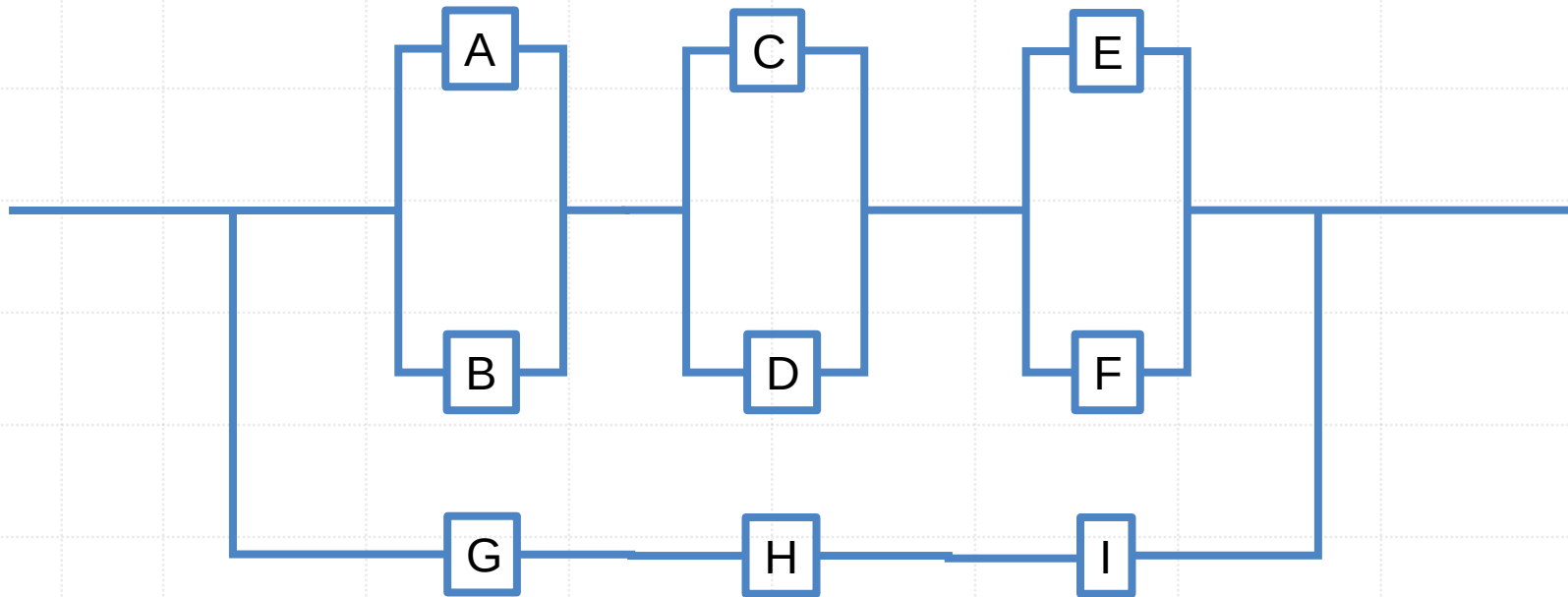


Create functional block diagrams

- A system works if $[(A \text{ or } B) \text{ AND } (C \text{ or } D) \text{ AND } (E \text{ or } F)] \text{ or } (G \text{ and } H \text{ and } I)$

Create functional block diagrams

- A system works if [(A or B) AND (C or D) AND (E or F)] or (G and H and I)





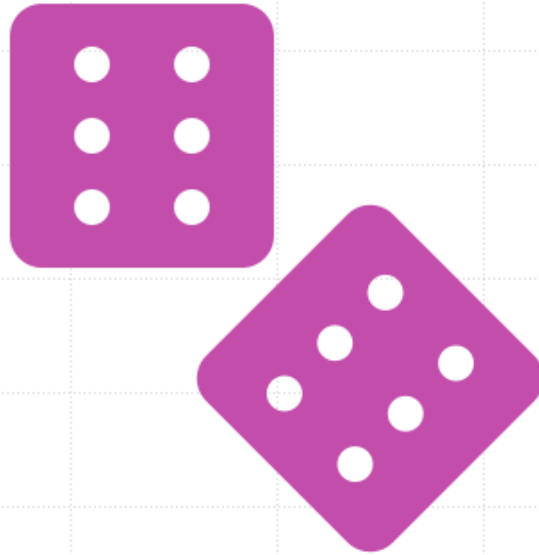
Functional Block Diagrams

- Functional Block Diagrams:
 - Represent functionality of a system or process
 - They are logic diagrams (AND and OR logic)
 - DO NOT provide the framework needed for probability calculations
- Need to convert to a system that takes probabilities
 - Failure Modes and Effects Analysis
 - Event Tree Analysis
 - Fault Tree Analysis



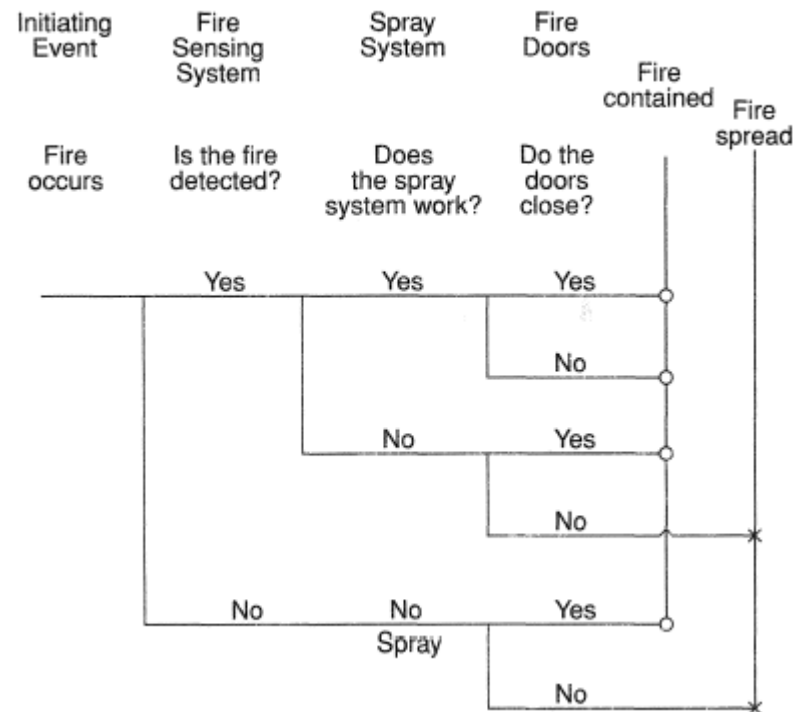
Effects of External Events

Probability of Failure



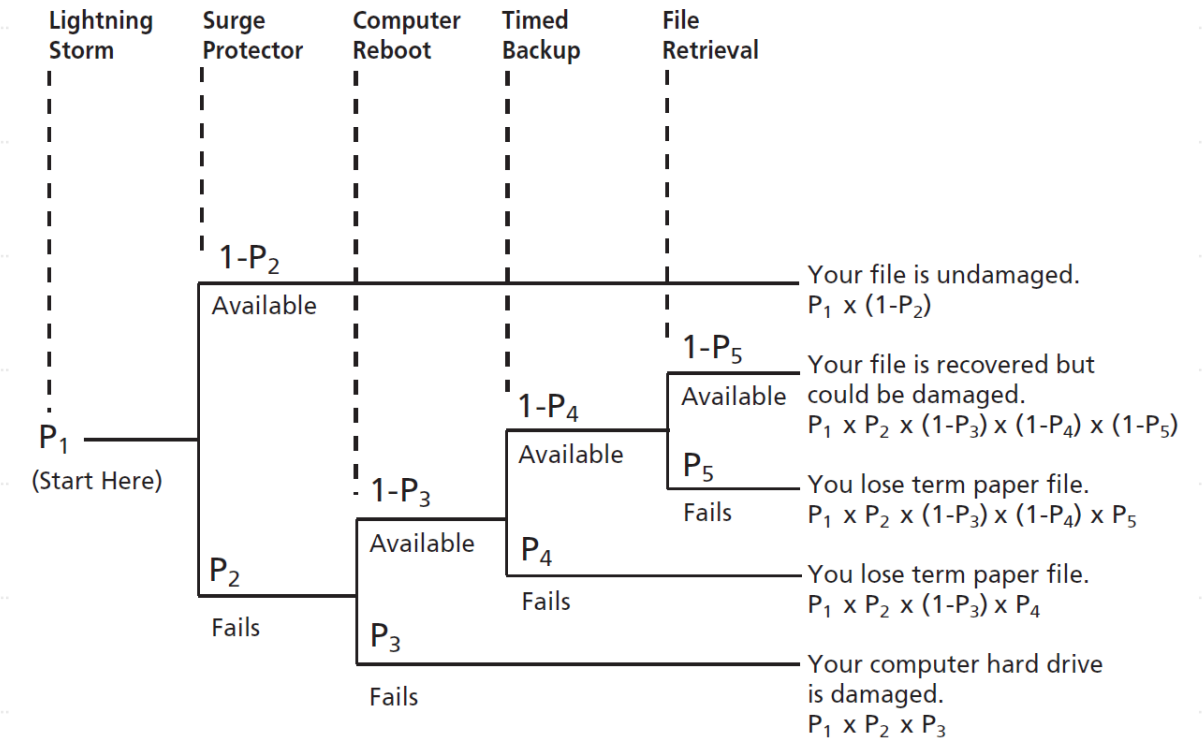
Event Tree Analysis

- Illustrates all possible sequences events after an initial event

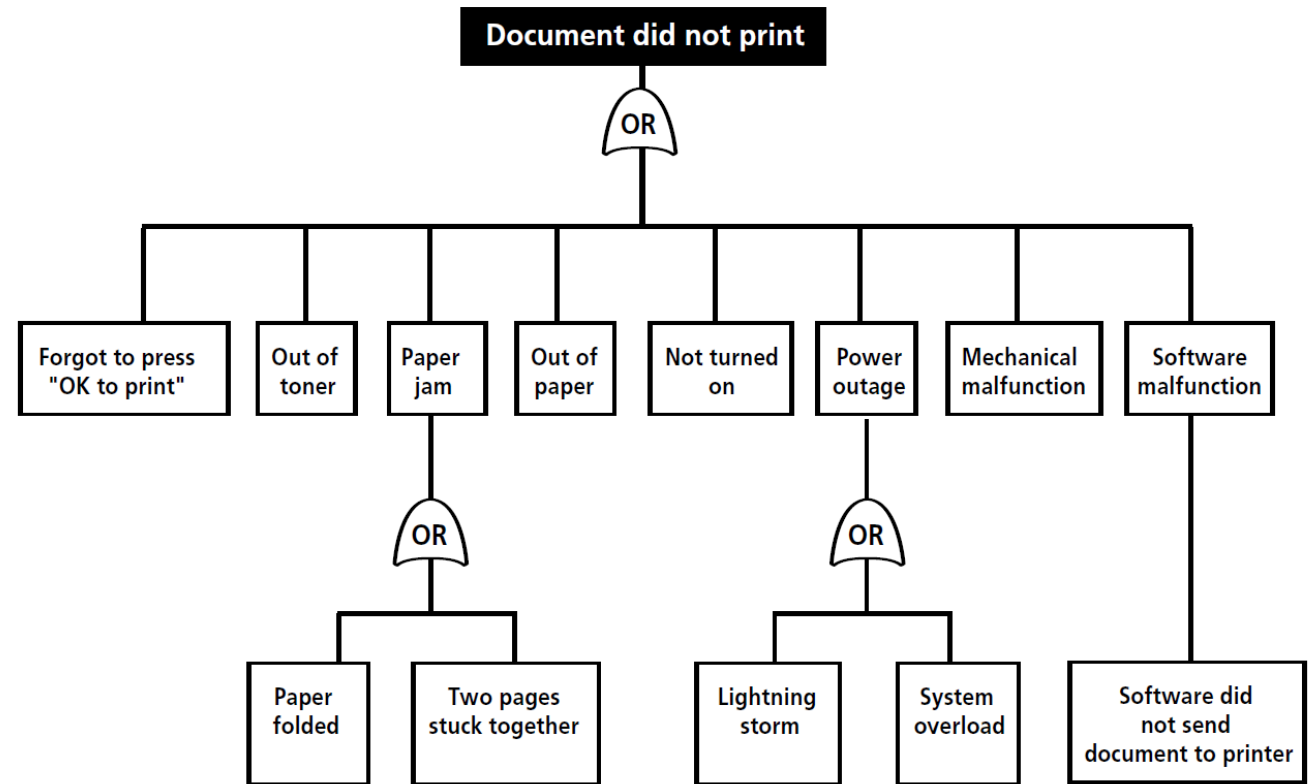


Risk Analysis Methods – Event Trees

*Losing Document due to
Lightning Storm*



Fault Trees





Fault Trees

- Top Down approach to understanding failures
- Identifies all possible contributors (Basic Events) that contribute to a specific outcome, the top level event
- Using Boolean logic to propagate failures (or probability of failure) from the basic level events to the top level events



Fault Trees

Relate the performance of a complex system to the performance of its individual components

Logic flows from bottom (components) to middle (sub-systems) to top (system)

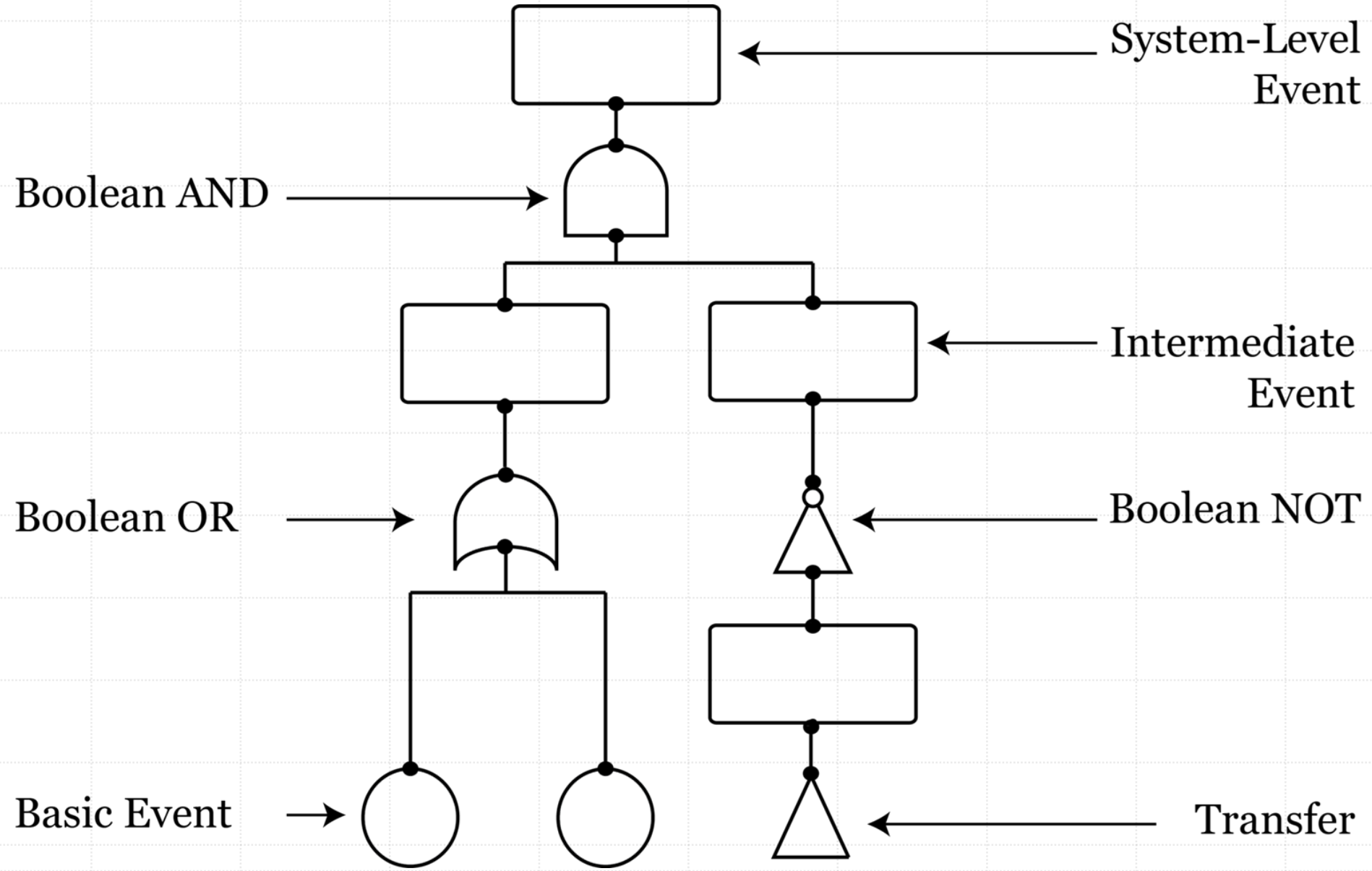
Related objects are connected by lines

Top-Down approach to understanding failures

Identifies all possible contributors (Basic Events) that contribute to a specific outcome, the top-level event

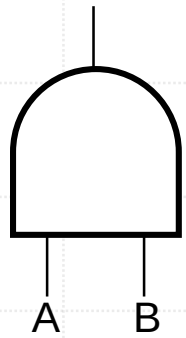
Using Boolean logic to propagate failures (or probability of failure) from the basic level events to the top-level events

Fault Tree Symbols

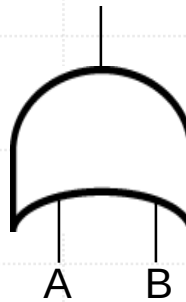


Fault Tree Symbols

Deterministic

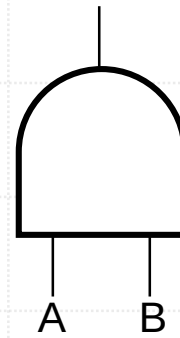


AND Gate: TRUE if **all** the events below the gate are TRUE

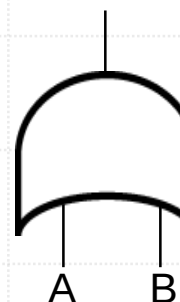


OR Gate: TRUE if **any** of the events below the gate are TRUE

Probabilistic



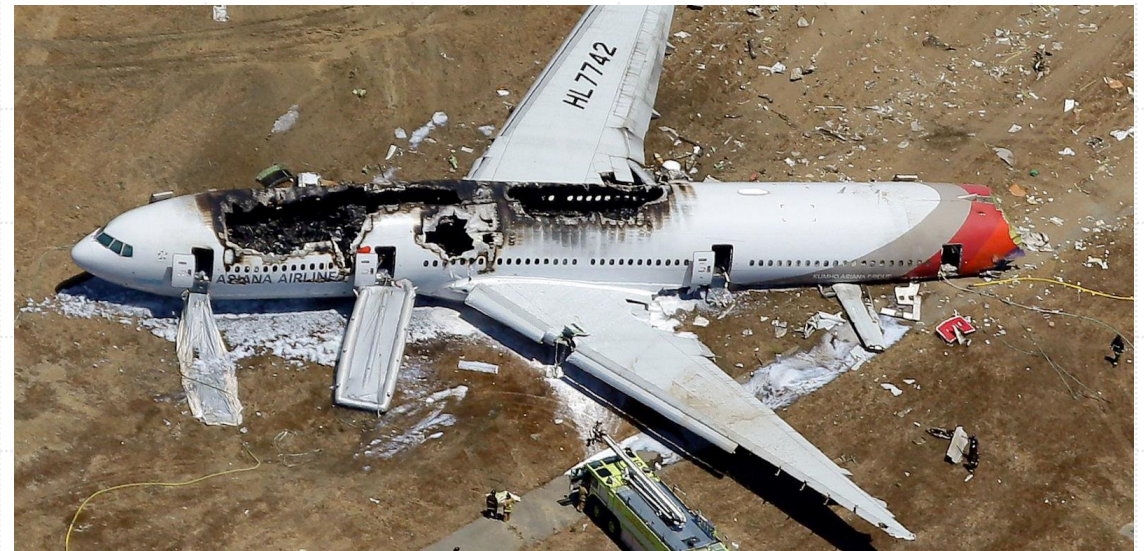
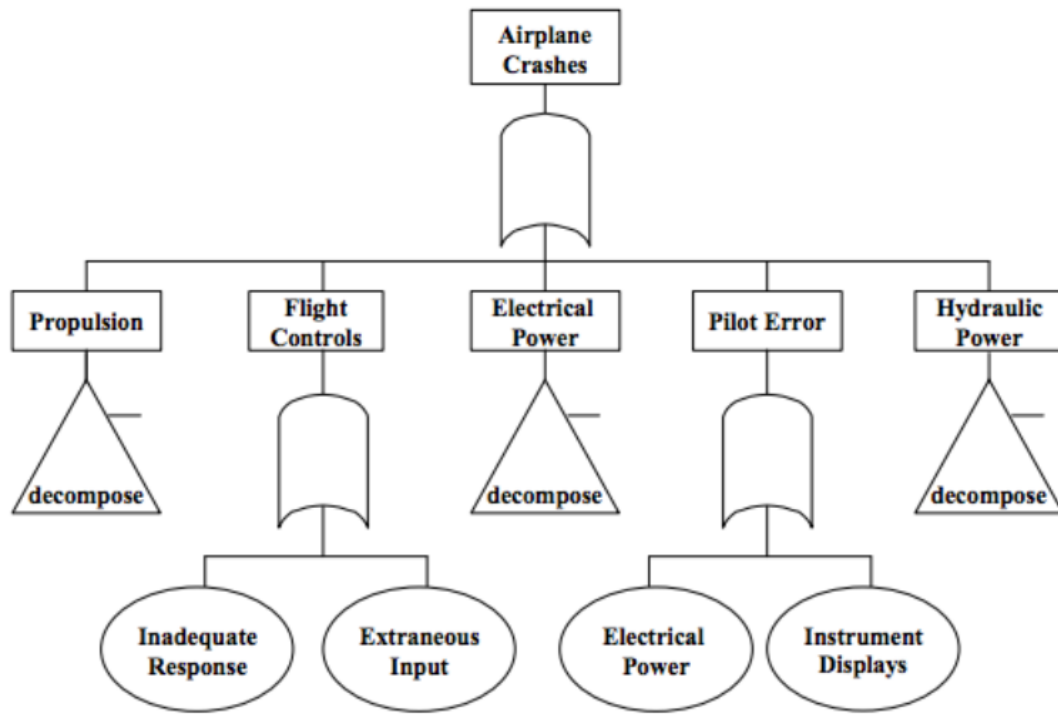
AND Gate: Probability of A and B
 $P(A \text{ and } B) = P(A)P(B)$



OR Gate: Probability of A or B
 $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$

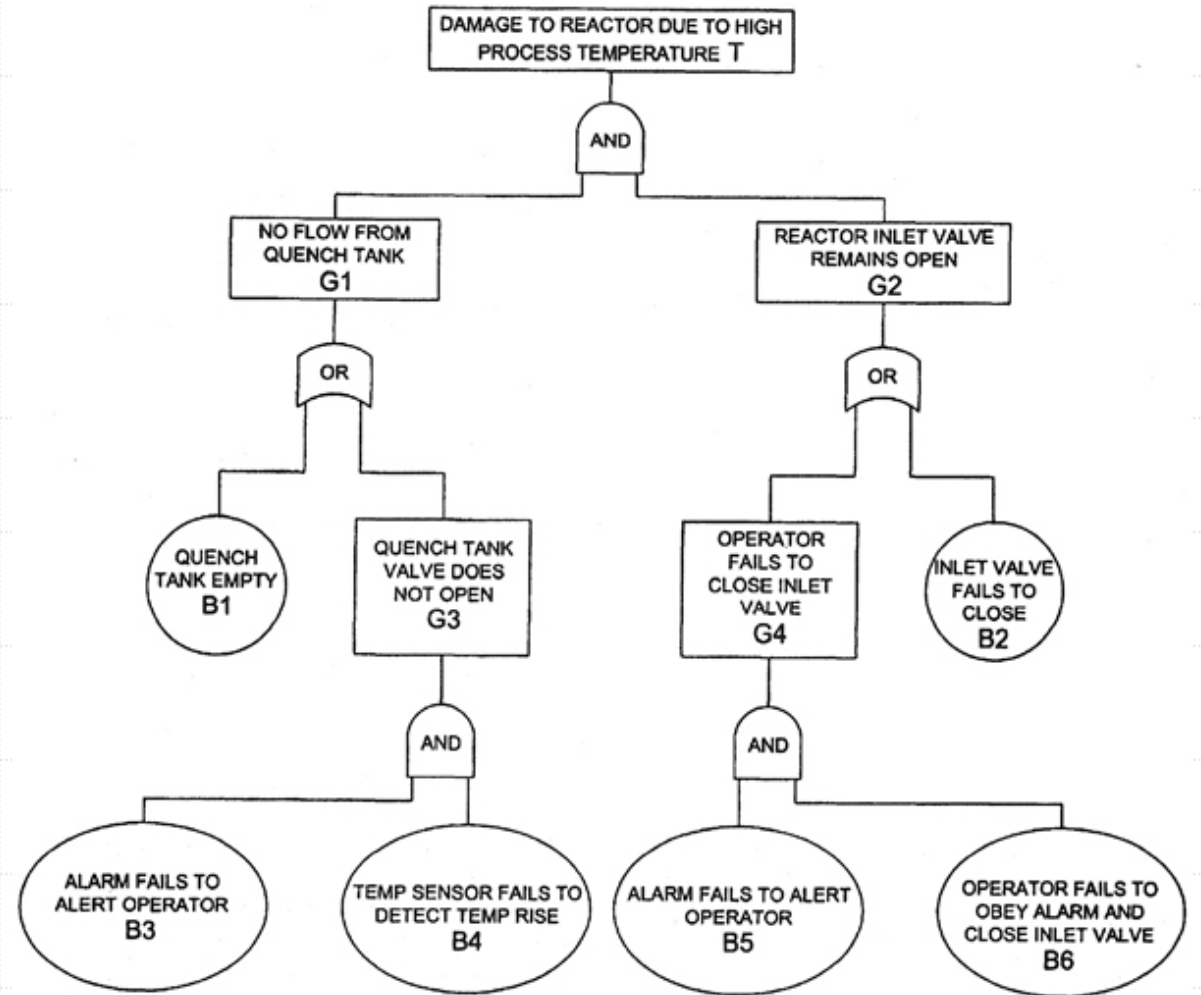
Fault Tree Examples

Airplane Crash



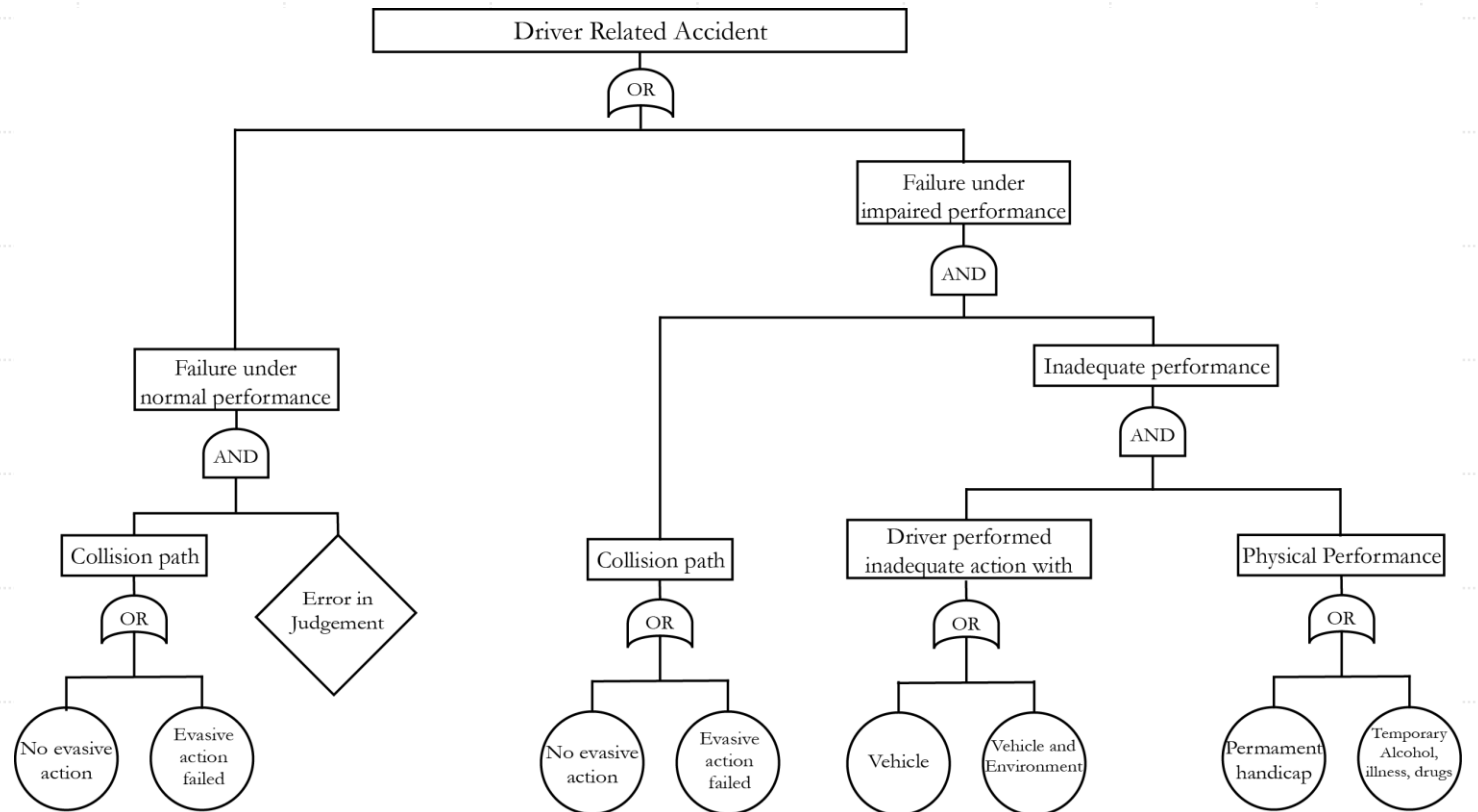
Fault tree examples

Nuclear Reactor Failure



Fault tree examples

Car Accident

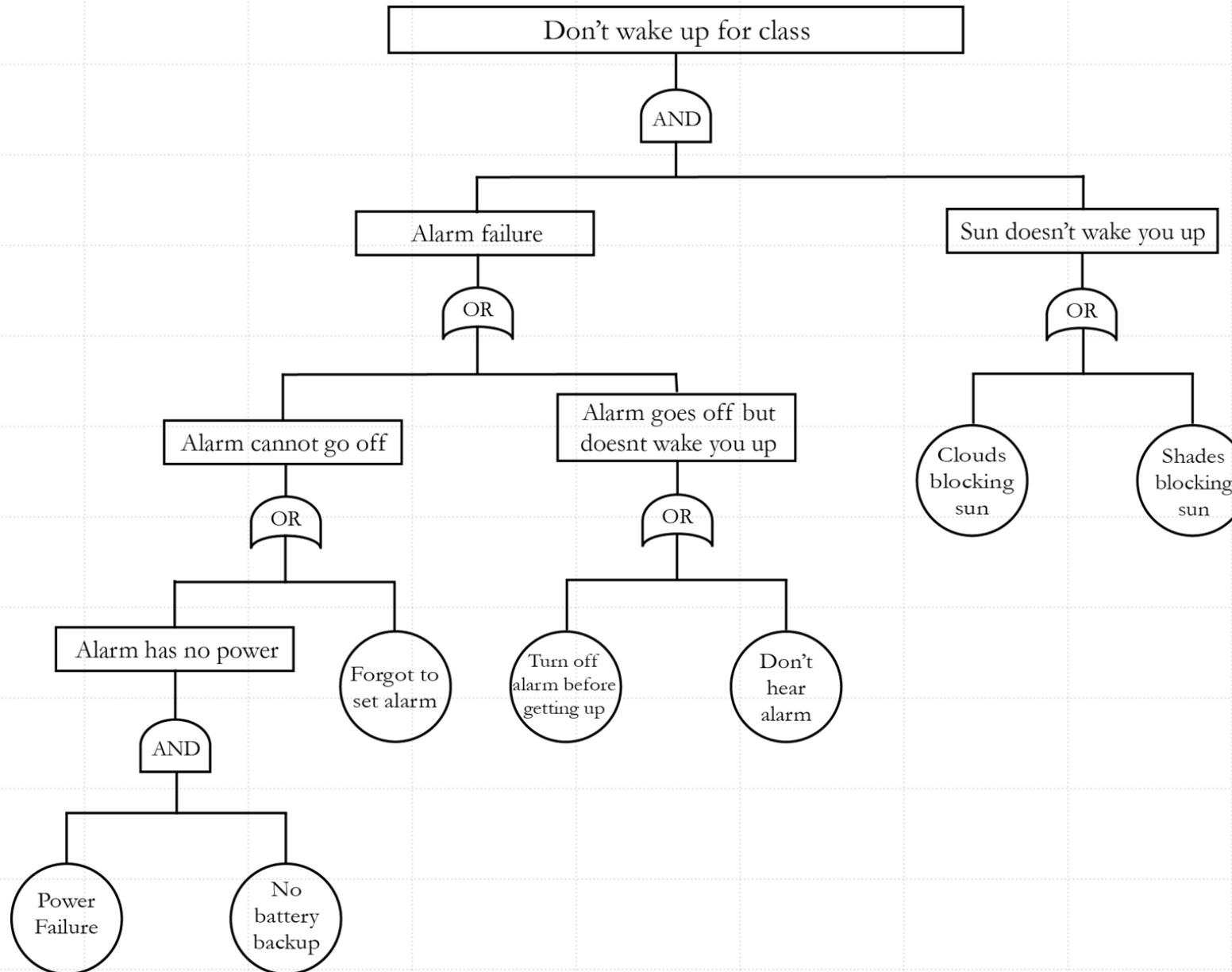




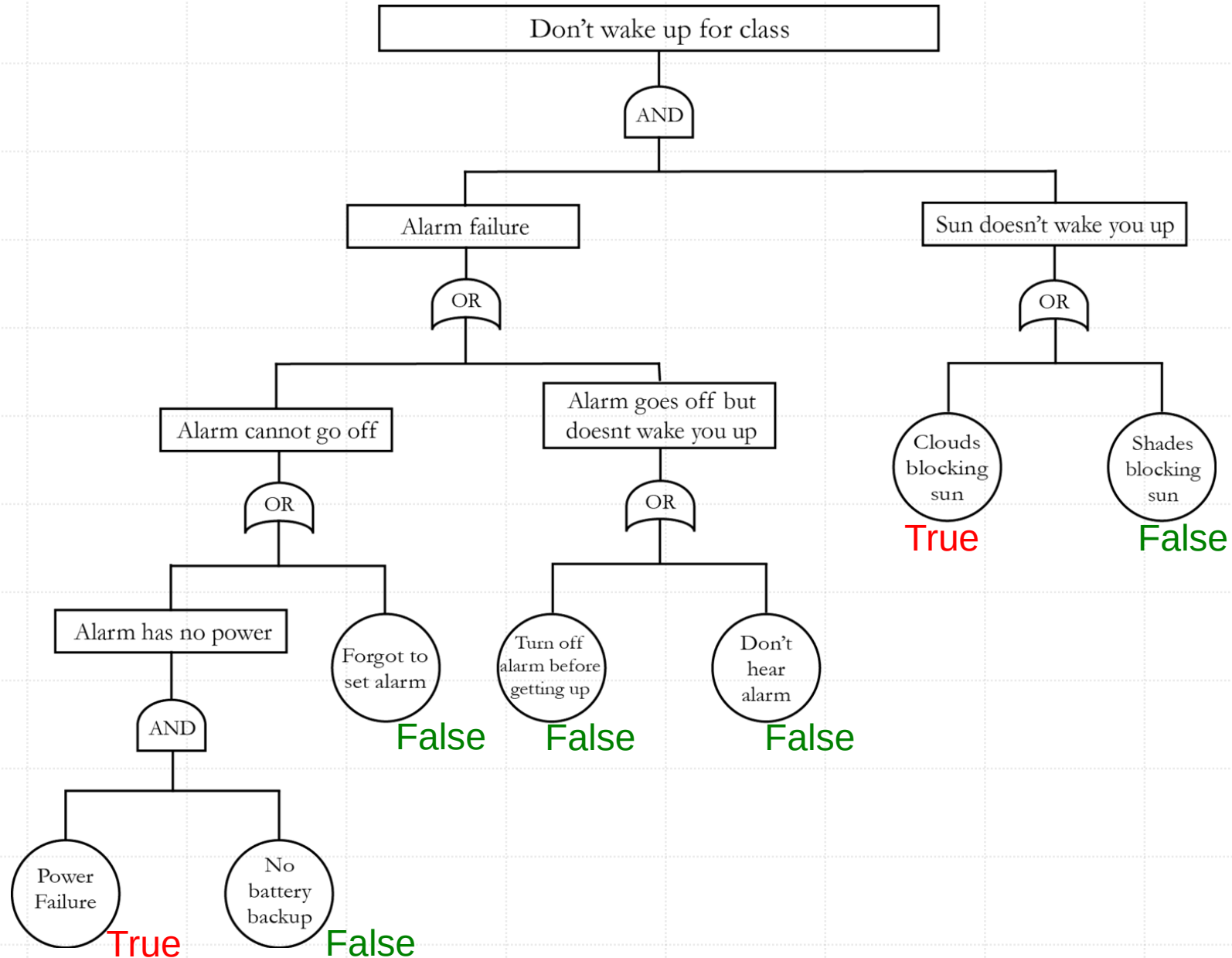
Create a Fault Tree

- Top level event: Don't wake up for class

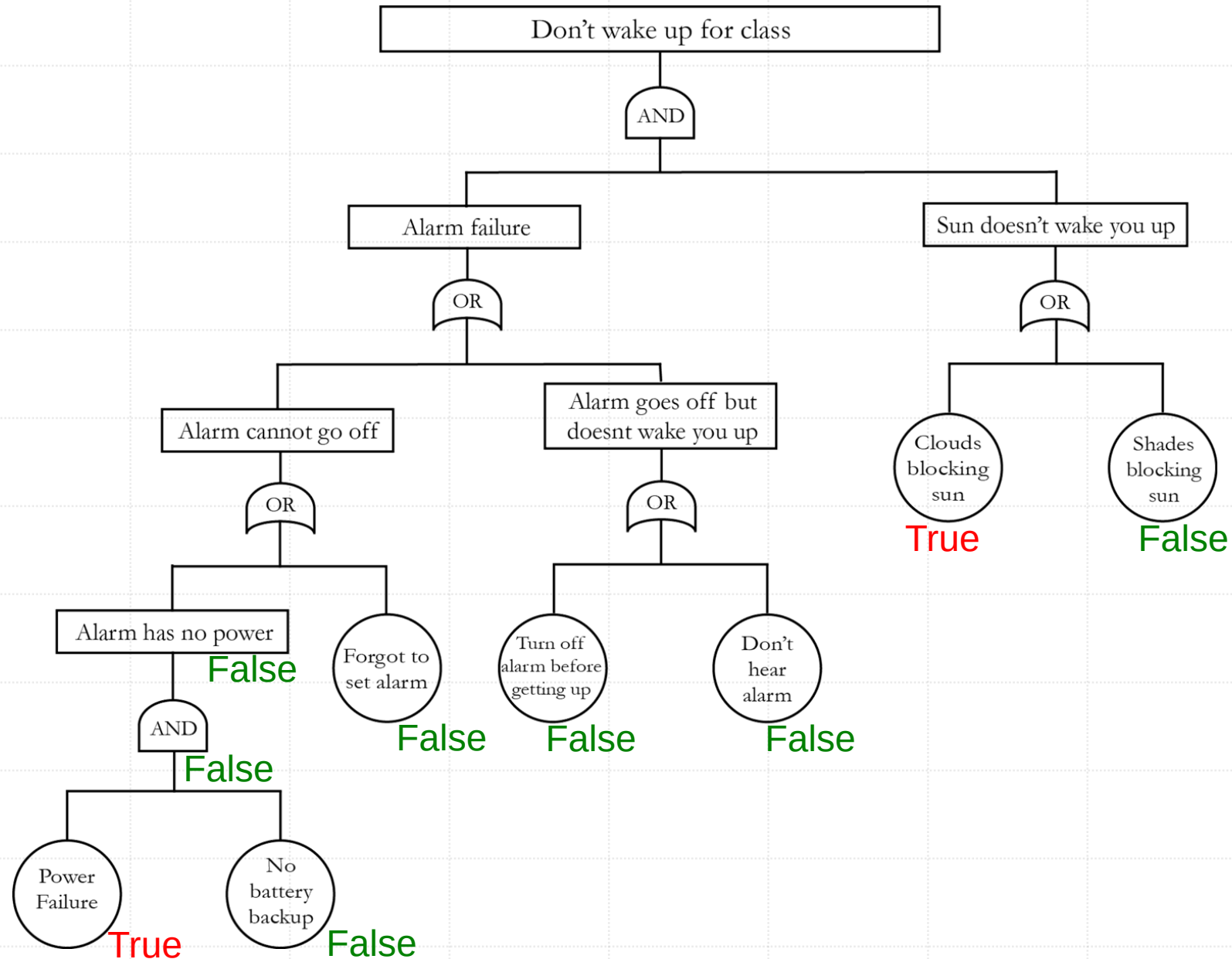
Deterministic Example



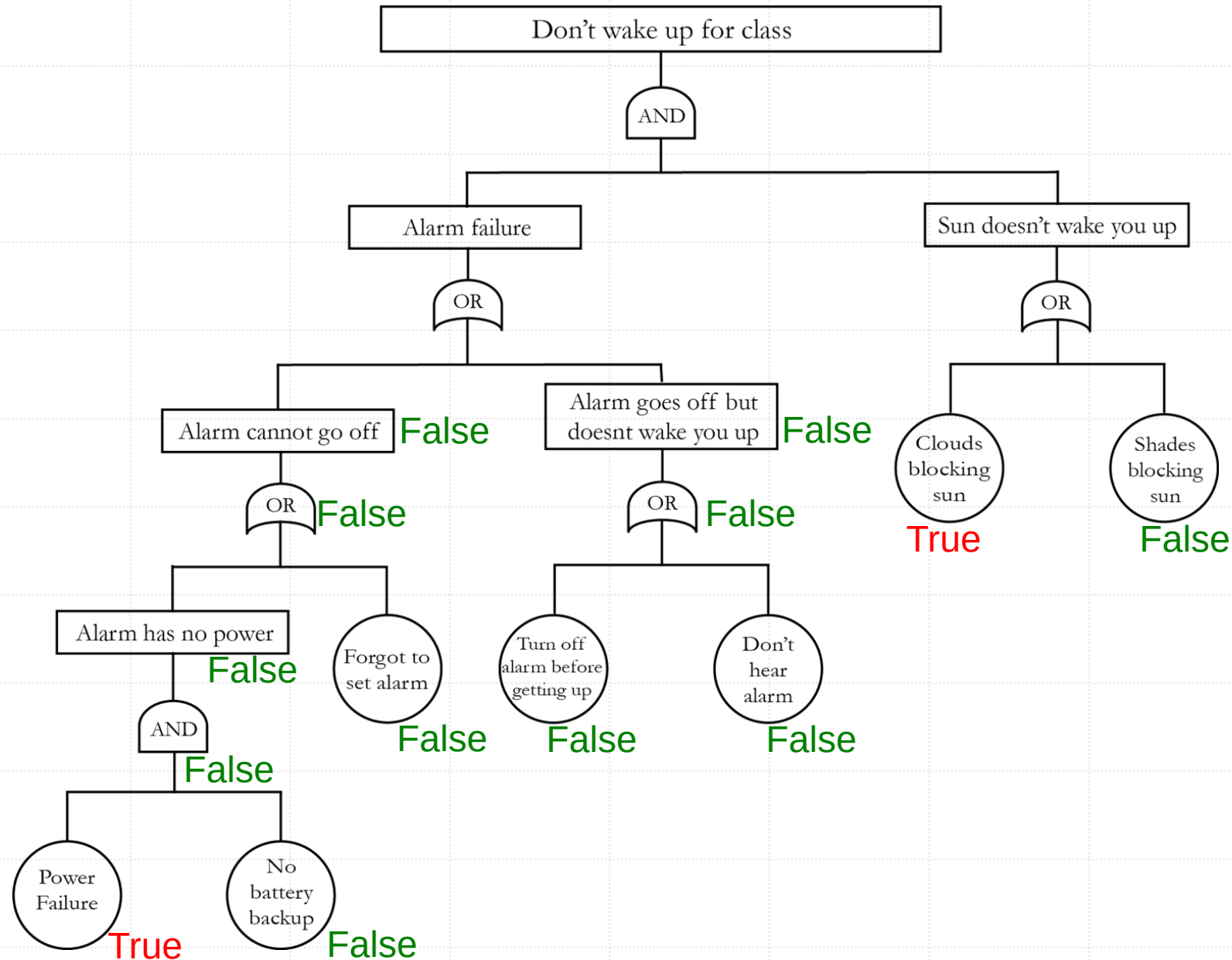
Deterministic Example



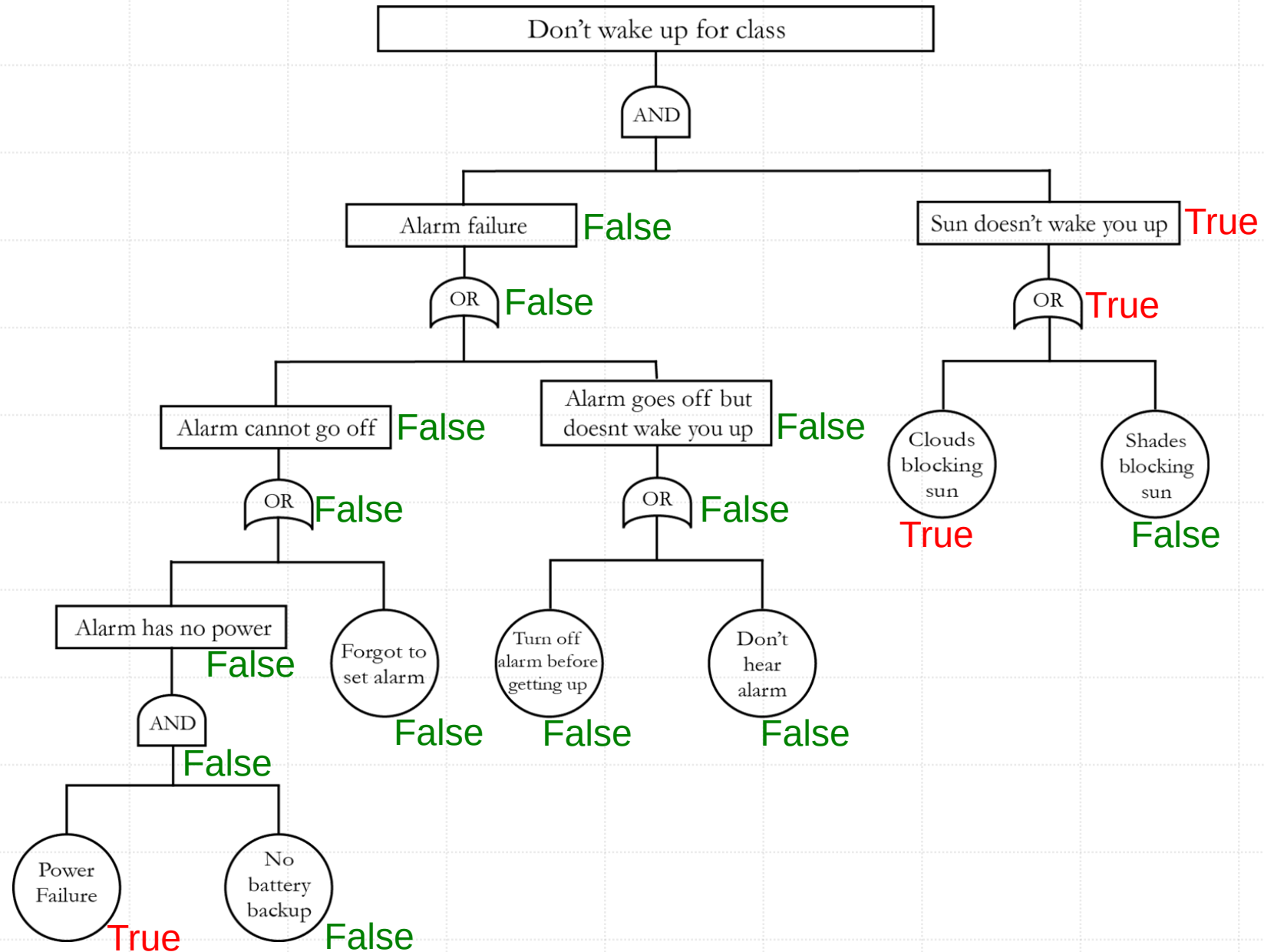
Deterministic Example



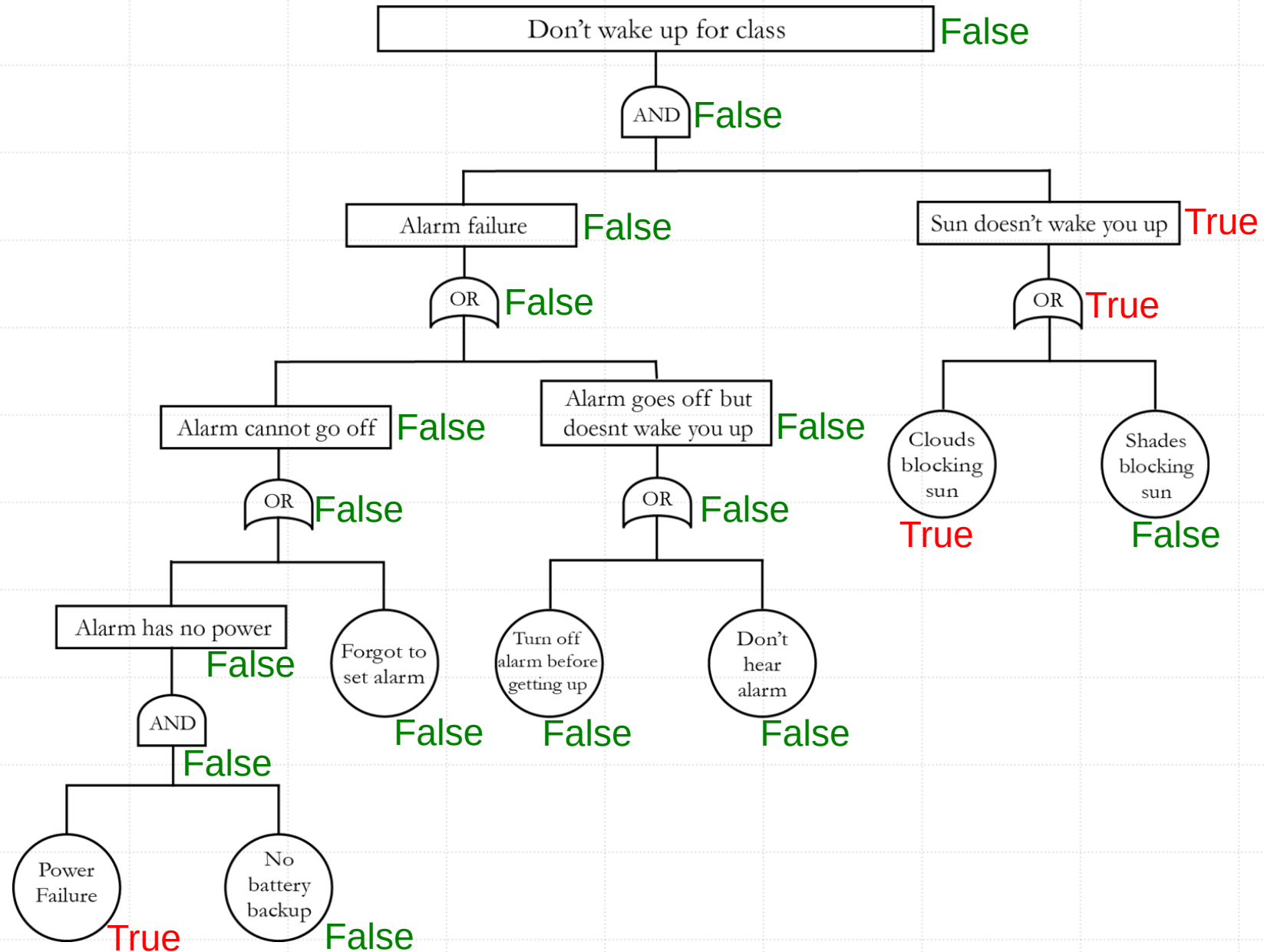
Deterministic Example



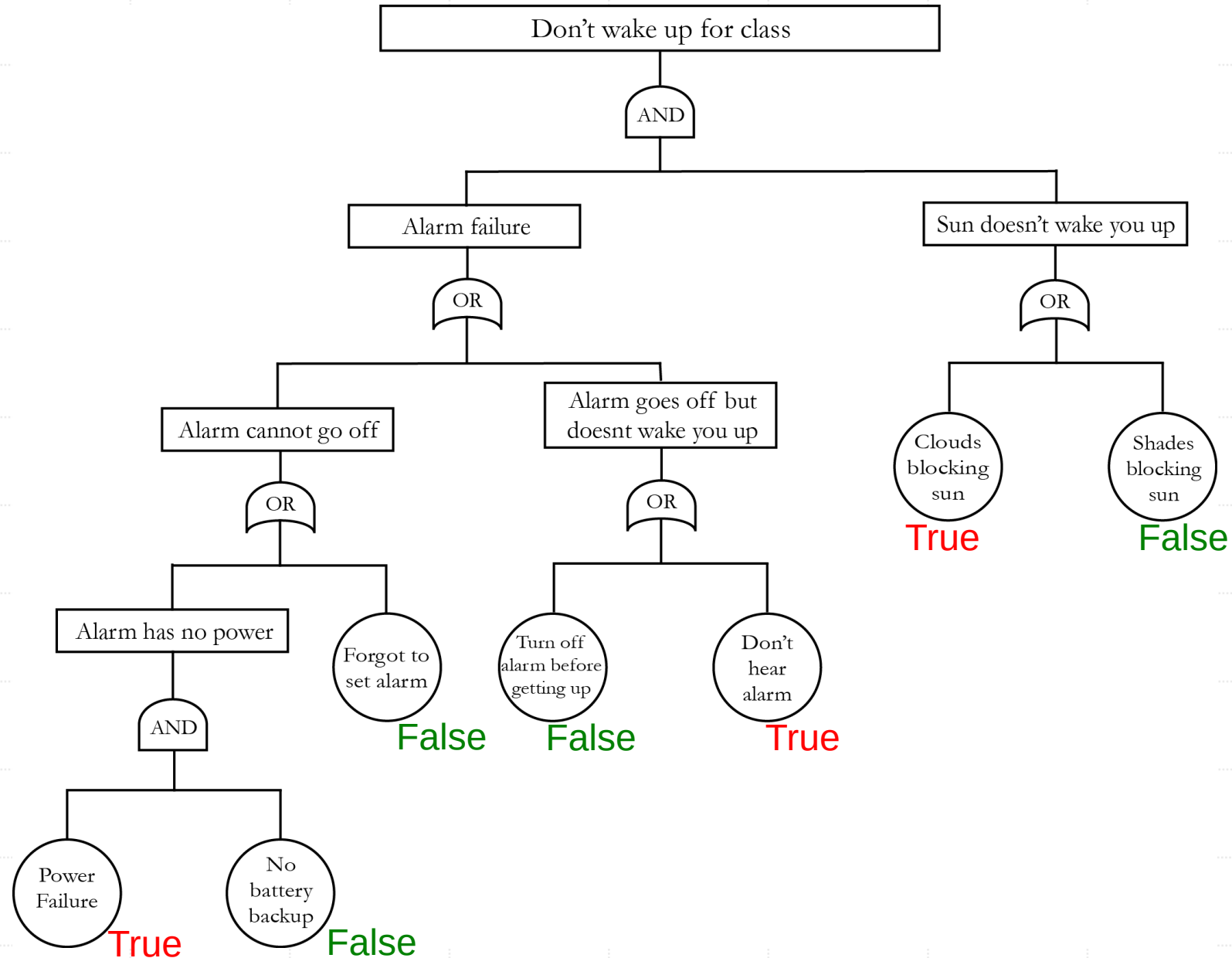
Deterministic Example



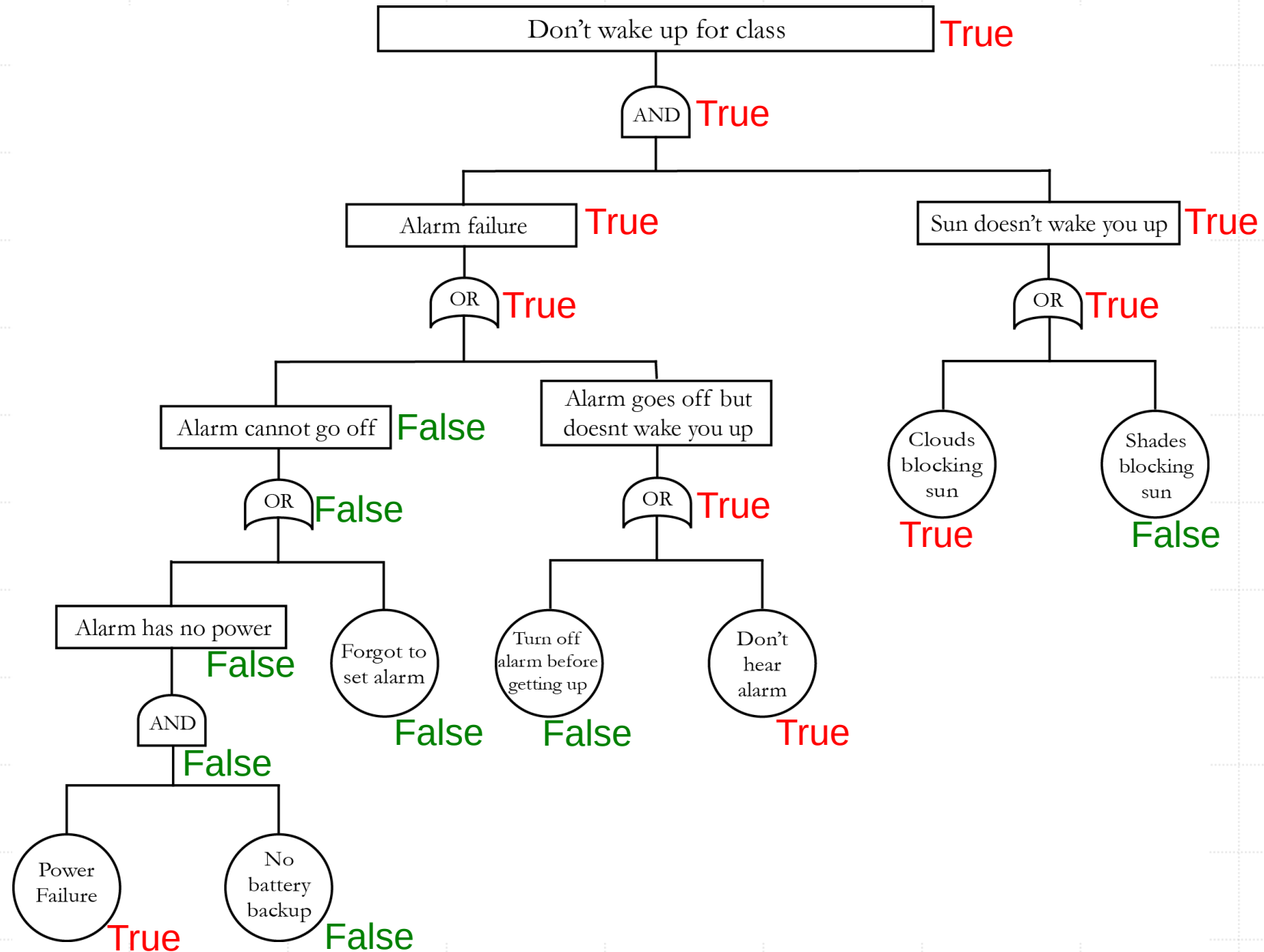
Deterministic Example



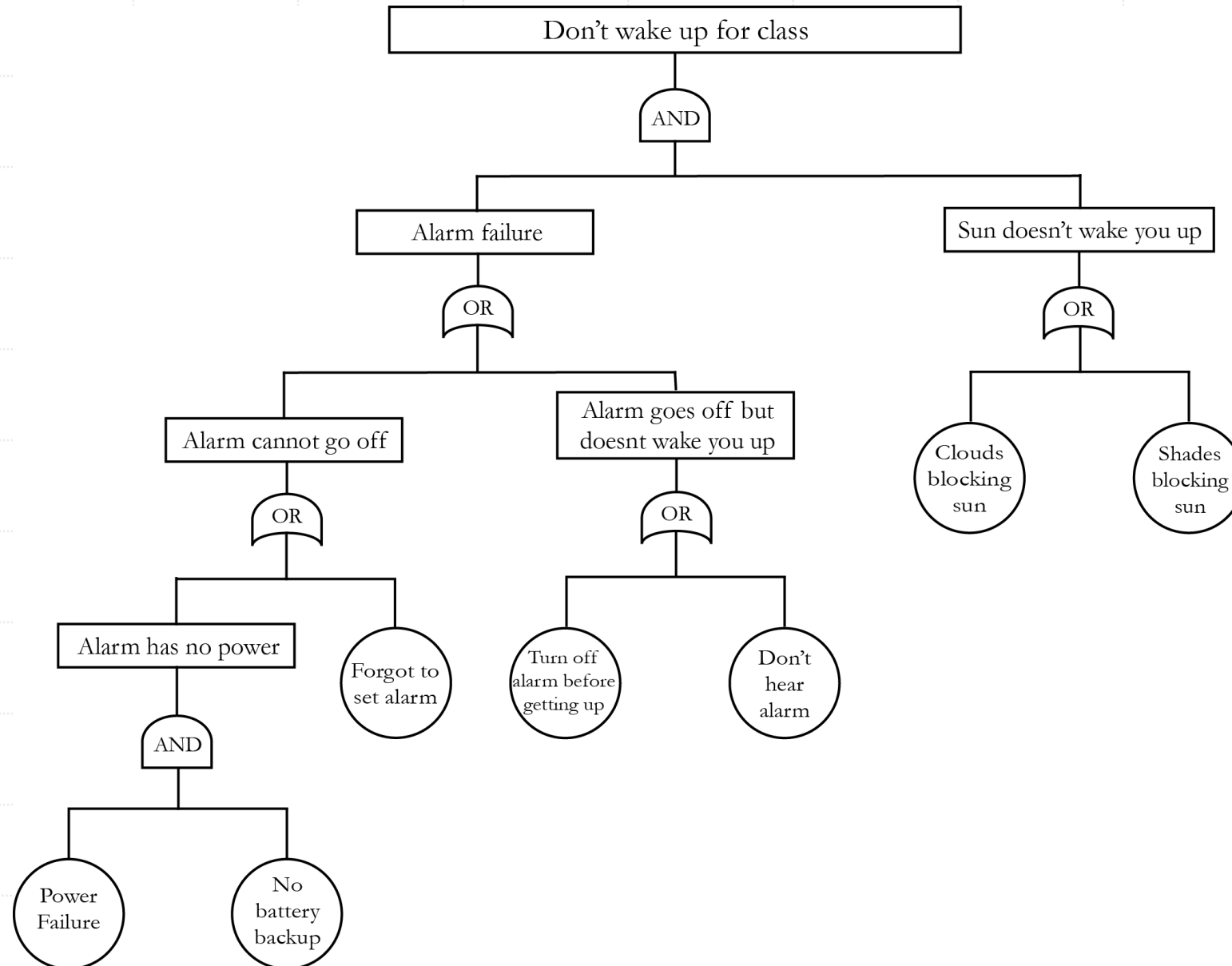
Deterministic Example



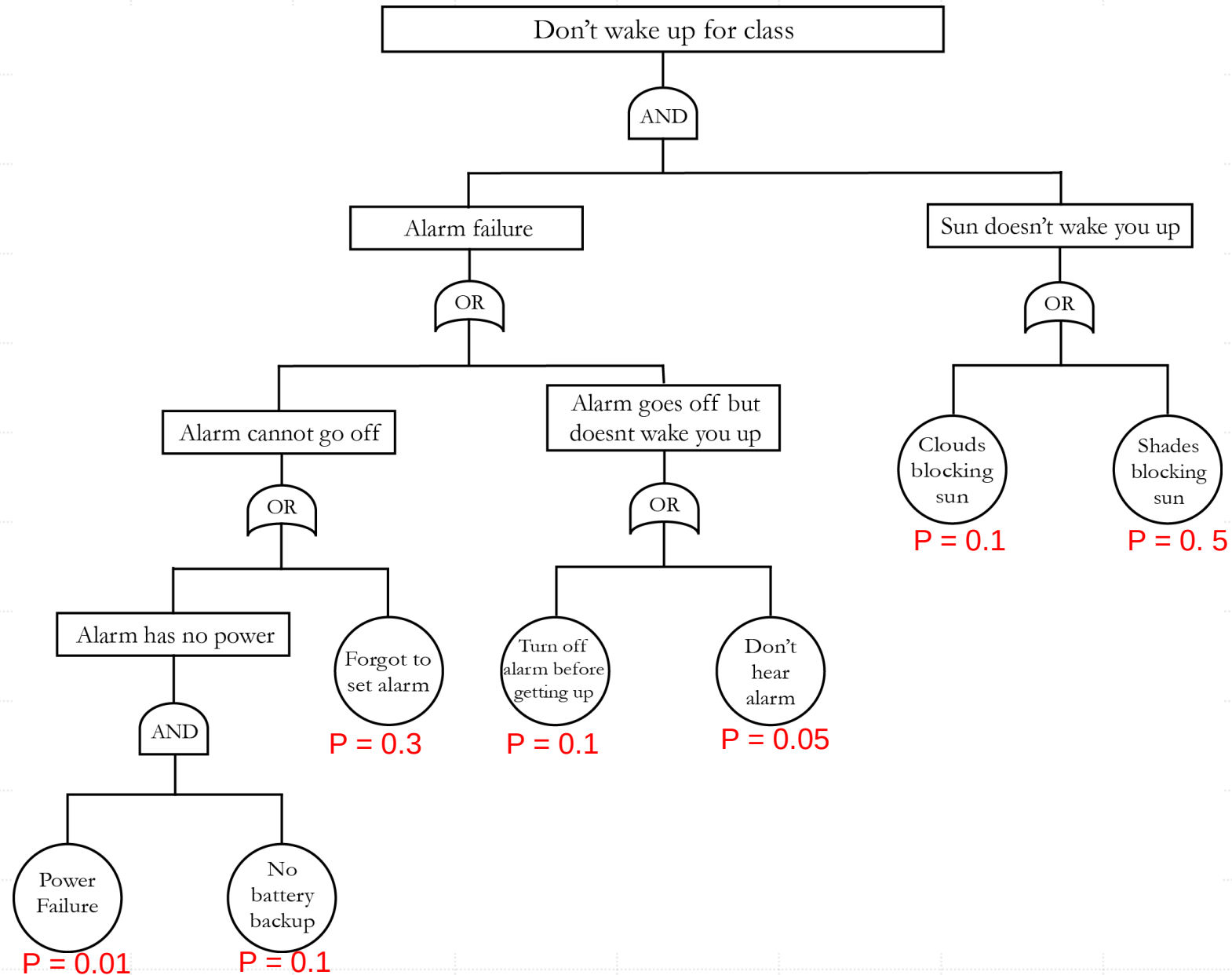
Deterministic Example



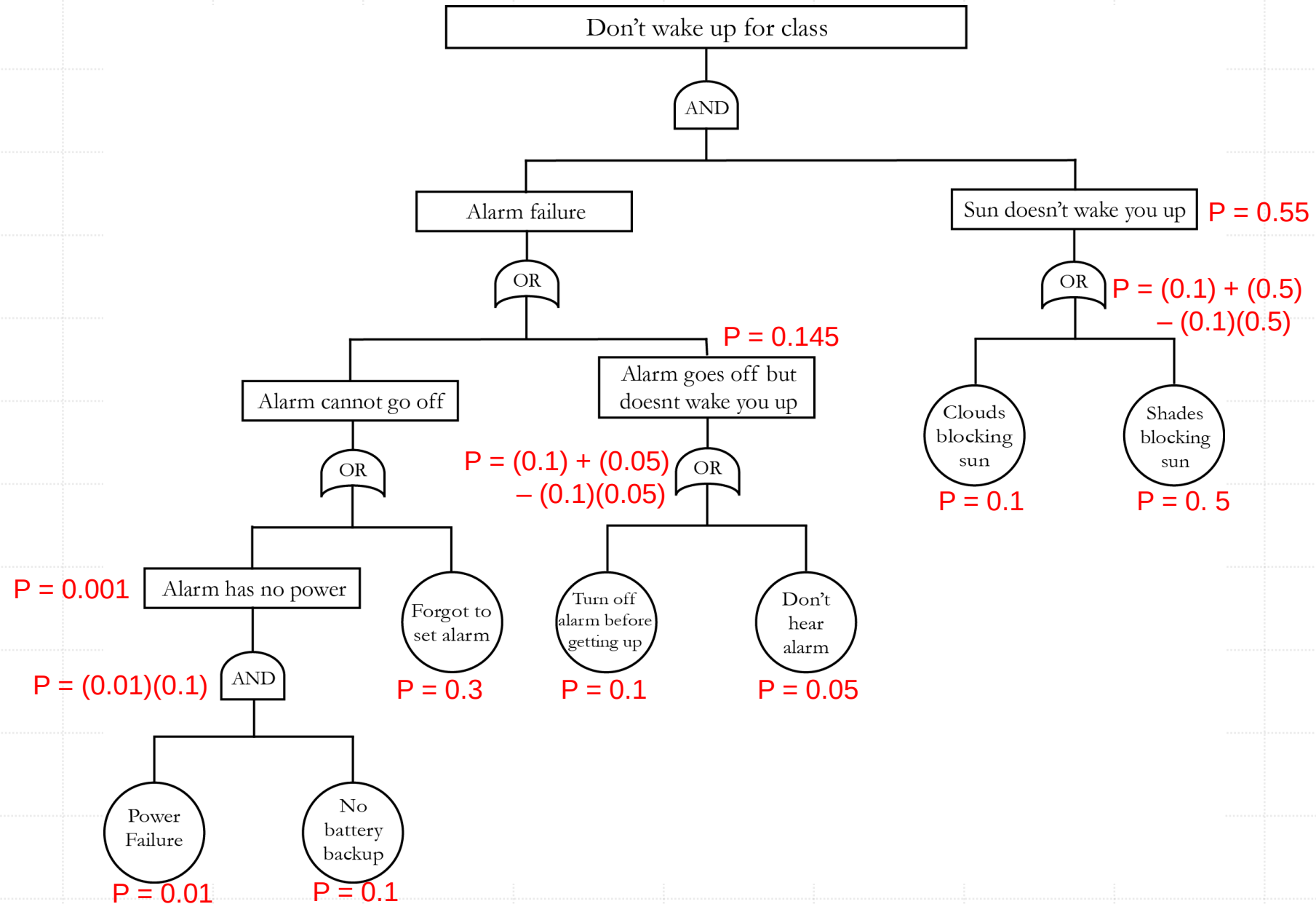
Probabilistic Example



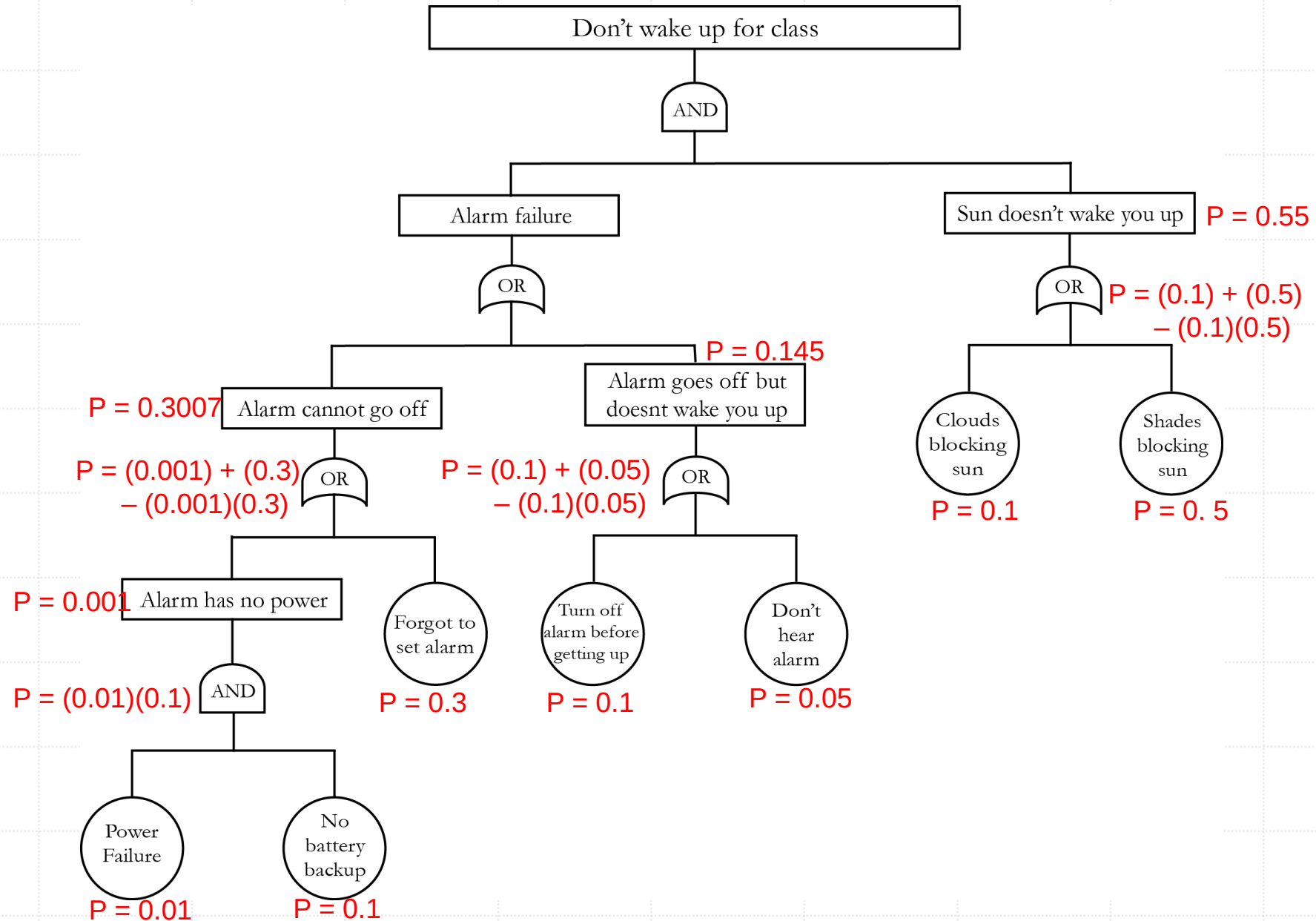
Probabilistic Example



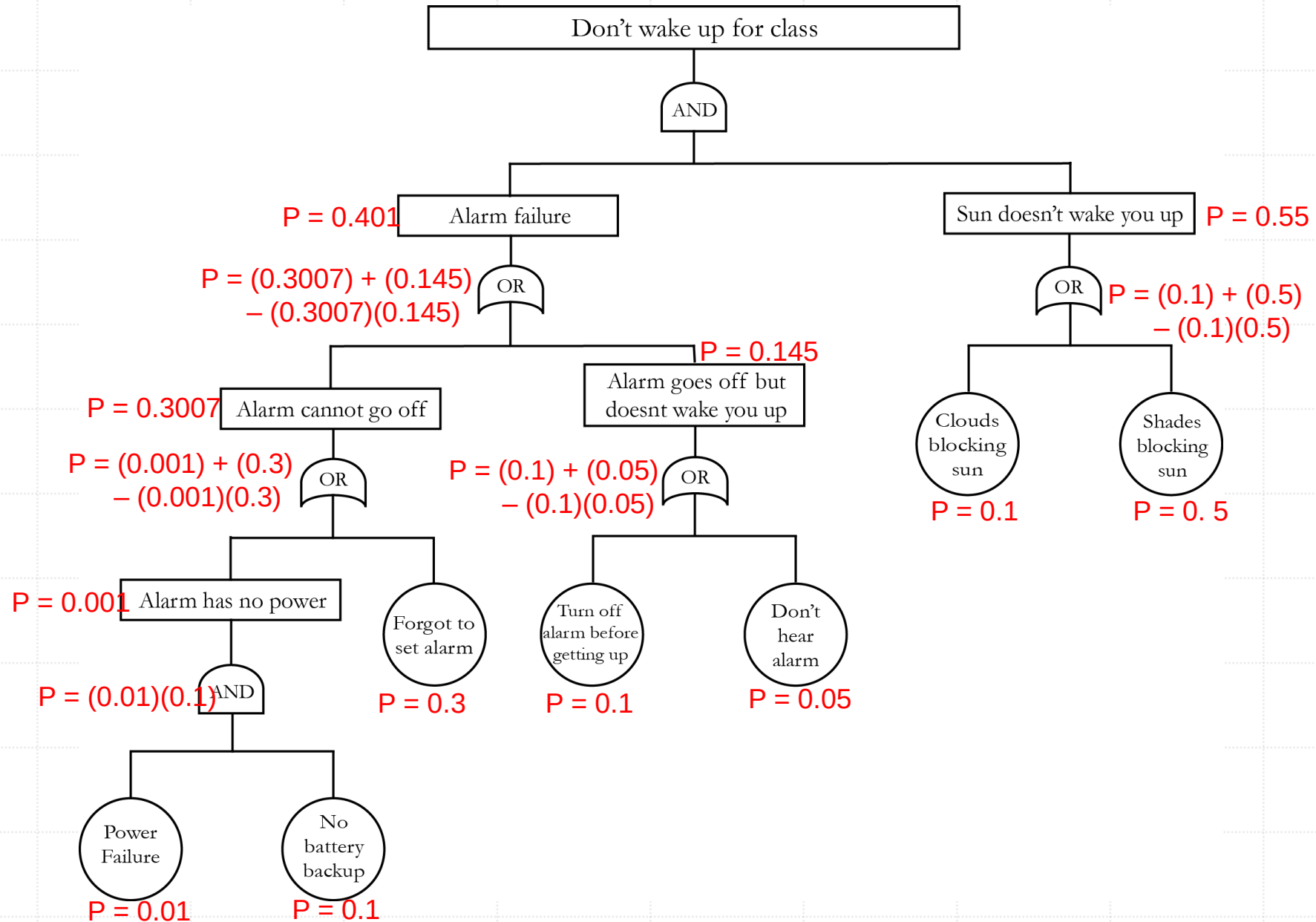
Probabilistic Example



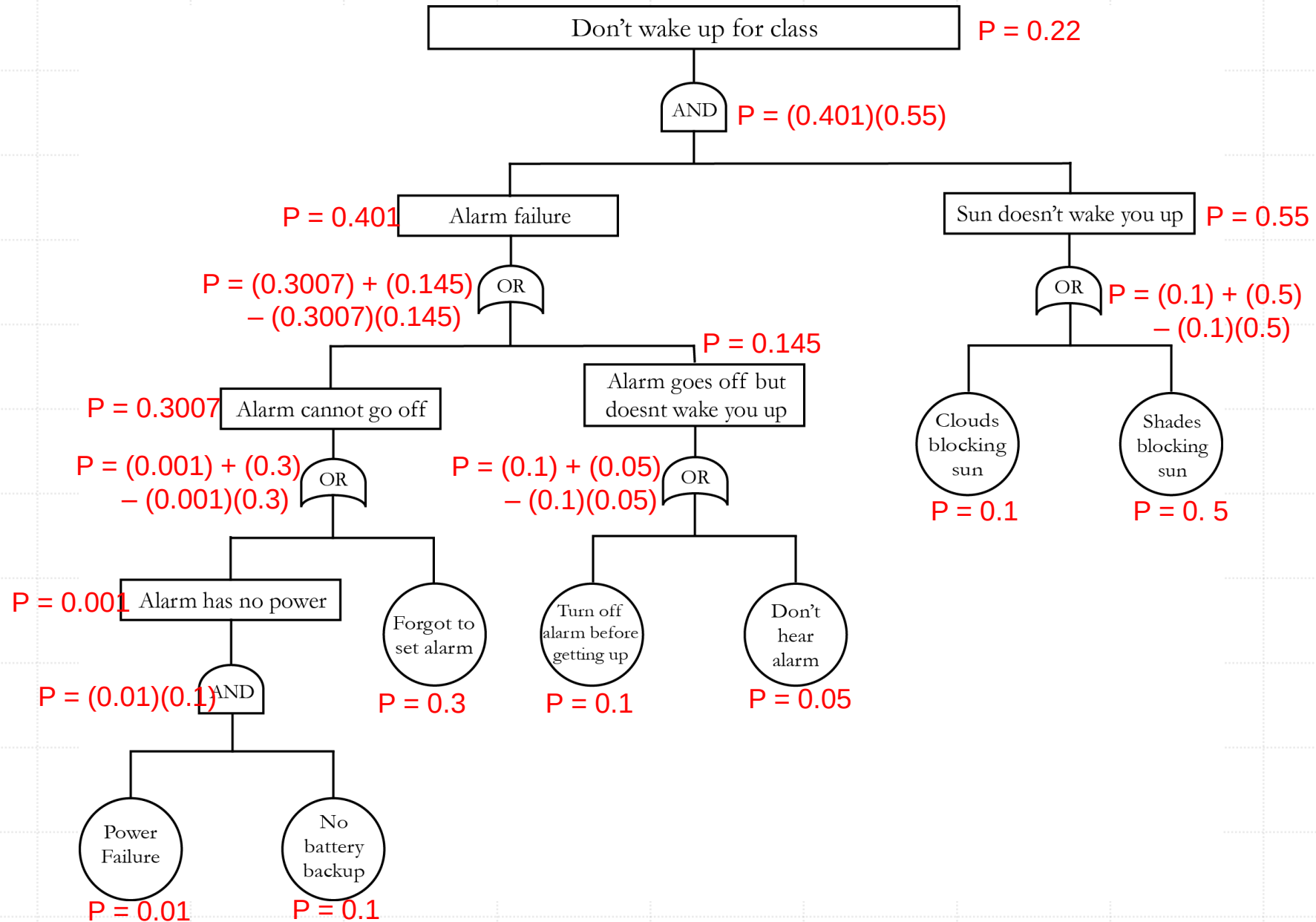
Probabilistic Example



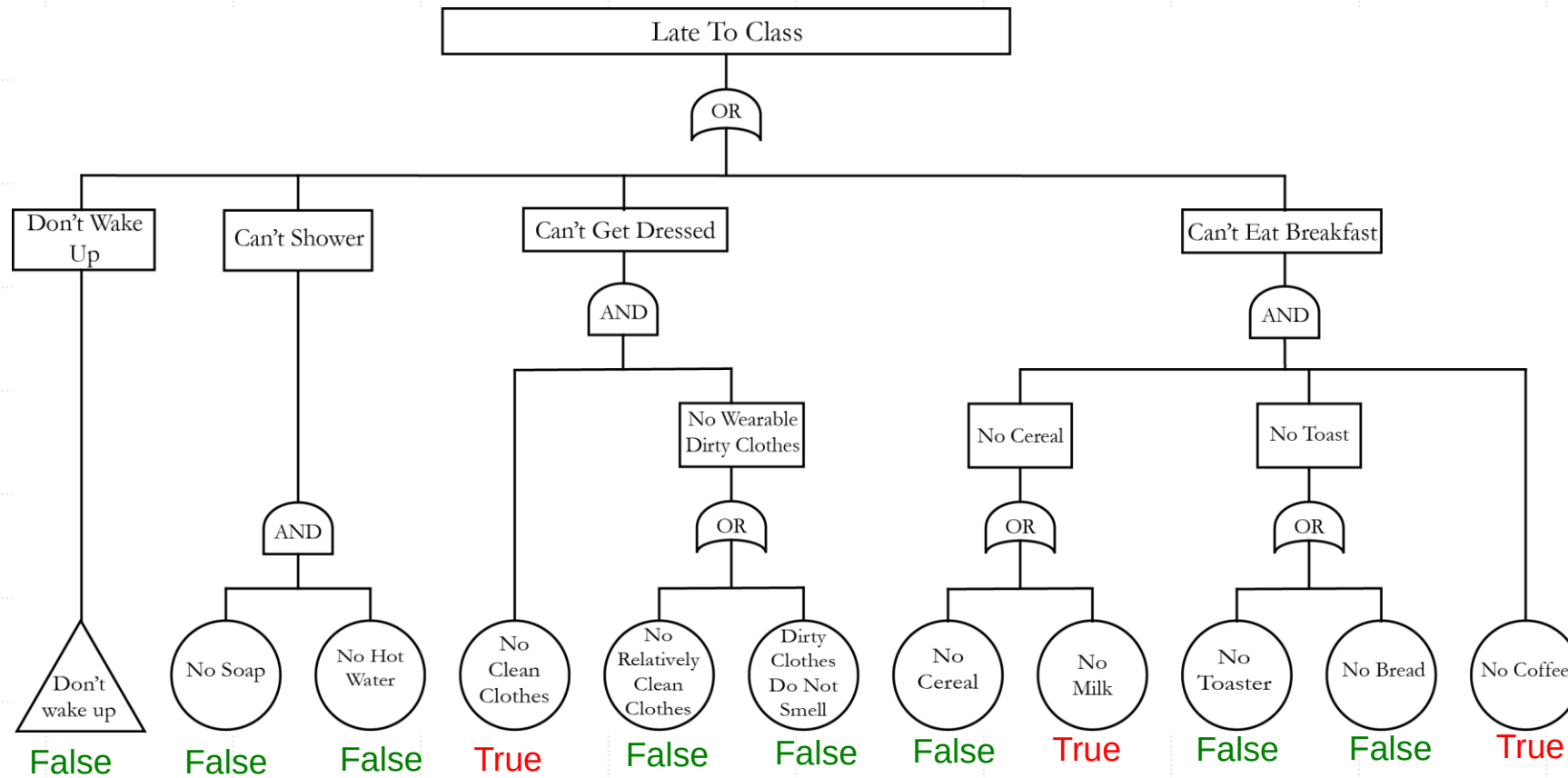
Probabilistic Example



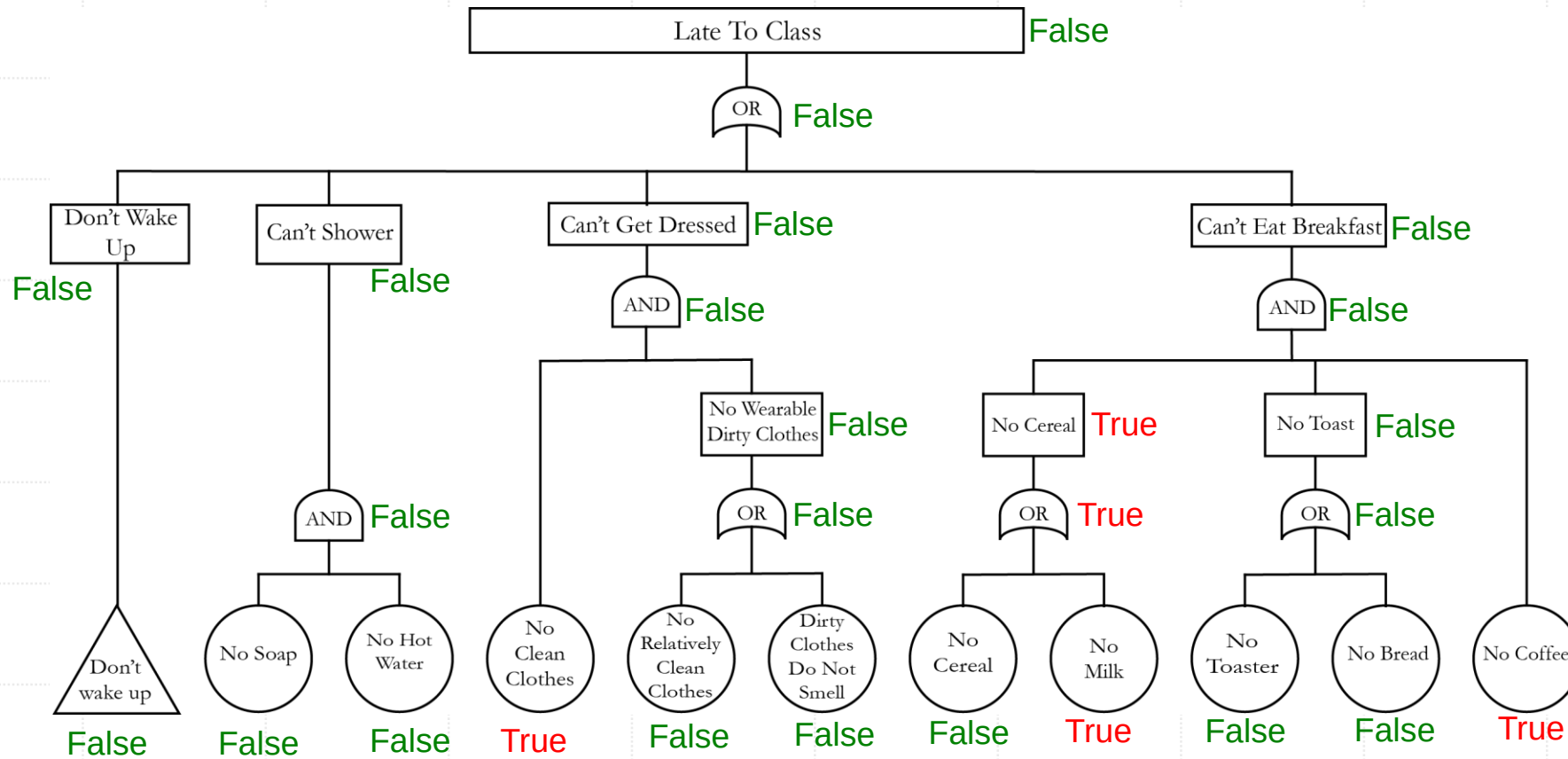
Probabilistic Example

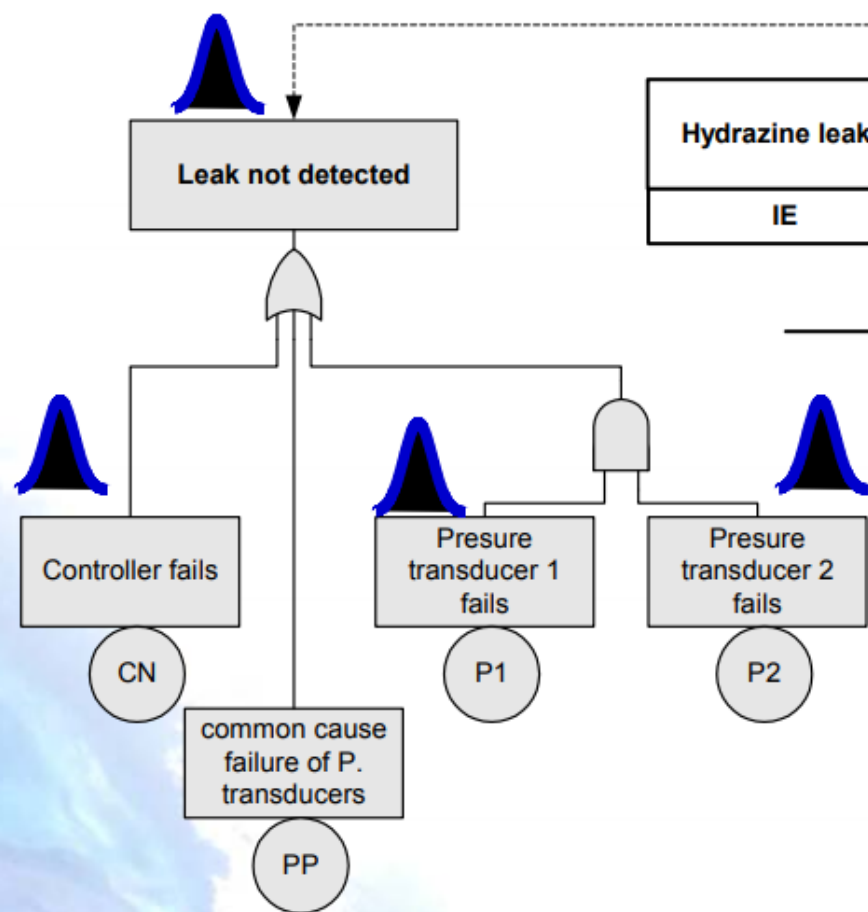


In Class Project



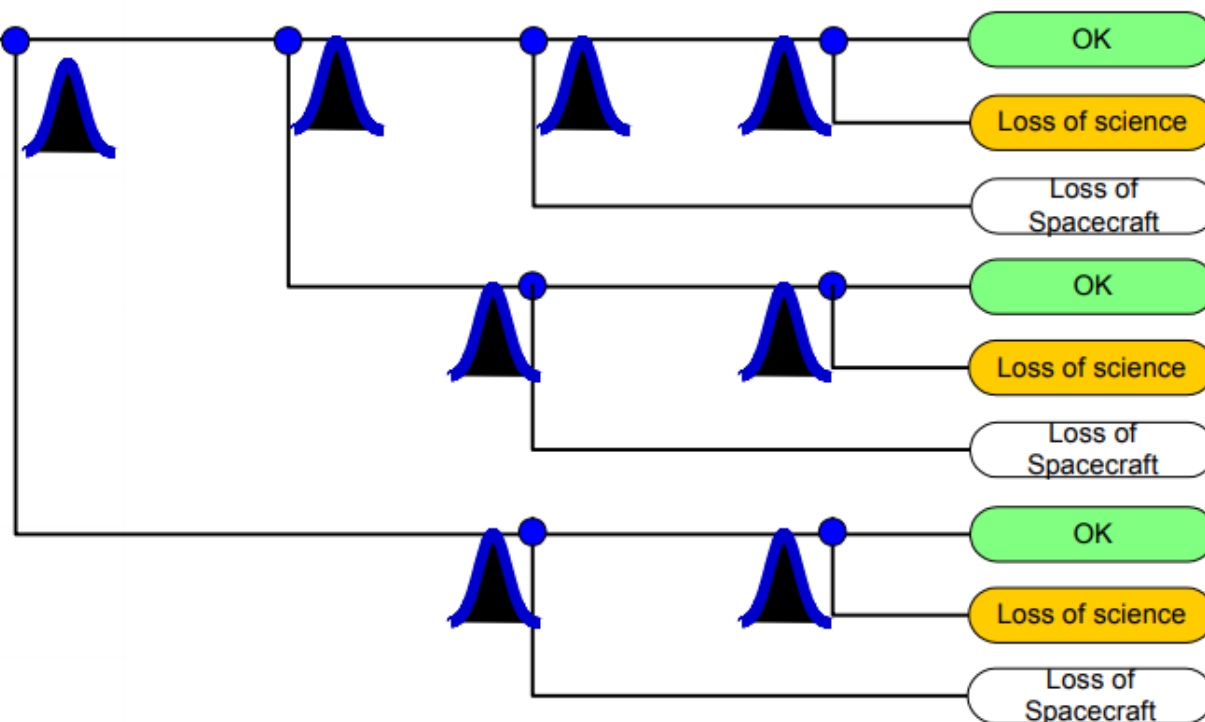
In Class Project Solution





Fault tree

Hydrazine leaks	Leak detected	Leak isolated	No damage to flight critical avionics	No damage to scientific equipment	End state
IE	LD	LI	A	S	



Event tree

■ Failure Modes and Effects

Risk Analysis Methods

Failure Modes and Effects

Process Step	Potential Failure Mode	Potential Failure Effect	SEV ¹	Potential Causes	OCC ²	Current Process Controls	DET ³	RPN ⁴	Action Recommended
What is the step?	In what ways can the step go wrong?	What is the impact on the customer if the failure mode is not prevented or corrected?	How severe is the effect on the customer?	What causes the step to go wrong (i.e., how could the failure mode occur)?	How frequently is the cause likely to occur?	What are the existing controls that either prevent the failure mode from occurring or detect it should it occur?	How probable is detection of the failure mode or its cause?	Risk priority number calculated as SEV x OCC x DET	What are the actions for reducing the occurrence of the cause or for improving its detection? Provide actions on all high RPNs and on severity ratings of 9 or 10.
ATM Pin Authentication	Unauthorized access	- Unauthorized cash withdrawal - Very dissatisfied customer	8	Lost or stolen ATM card	3	Block ATM card after three failed authentication attempts	3	72	
	Authentication failure	Annoyed customer	3	Network failure	5	Install load balancer to distribute work-load across network links	5	75	
Dispense Cash	Cash not disbursed	Dissatisfied customer	7	ATM out of cash	7	Internal alert of low cash in ATM	4	196	Increase minimum cash threshold limit of heavily used ATMs to prevent out-of-cash instances
	Account debited but no cash disbursed	Very dissatisfied customer	8	- Transaction failure - Network issue	3	Install load balancer to distribute work-load across network links	4	96	
	Extra cash dispensed	Bank loses money	8	- Bills stuck to each other - Bills stacked incorrectly	2	Verification while loading cash in ATM	3	48	

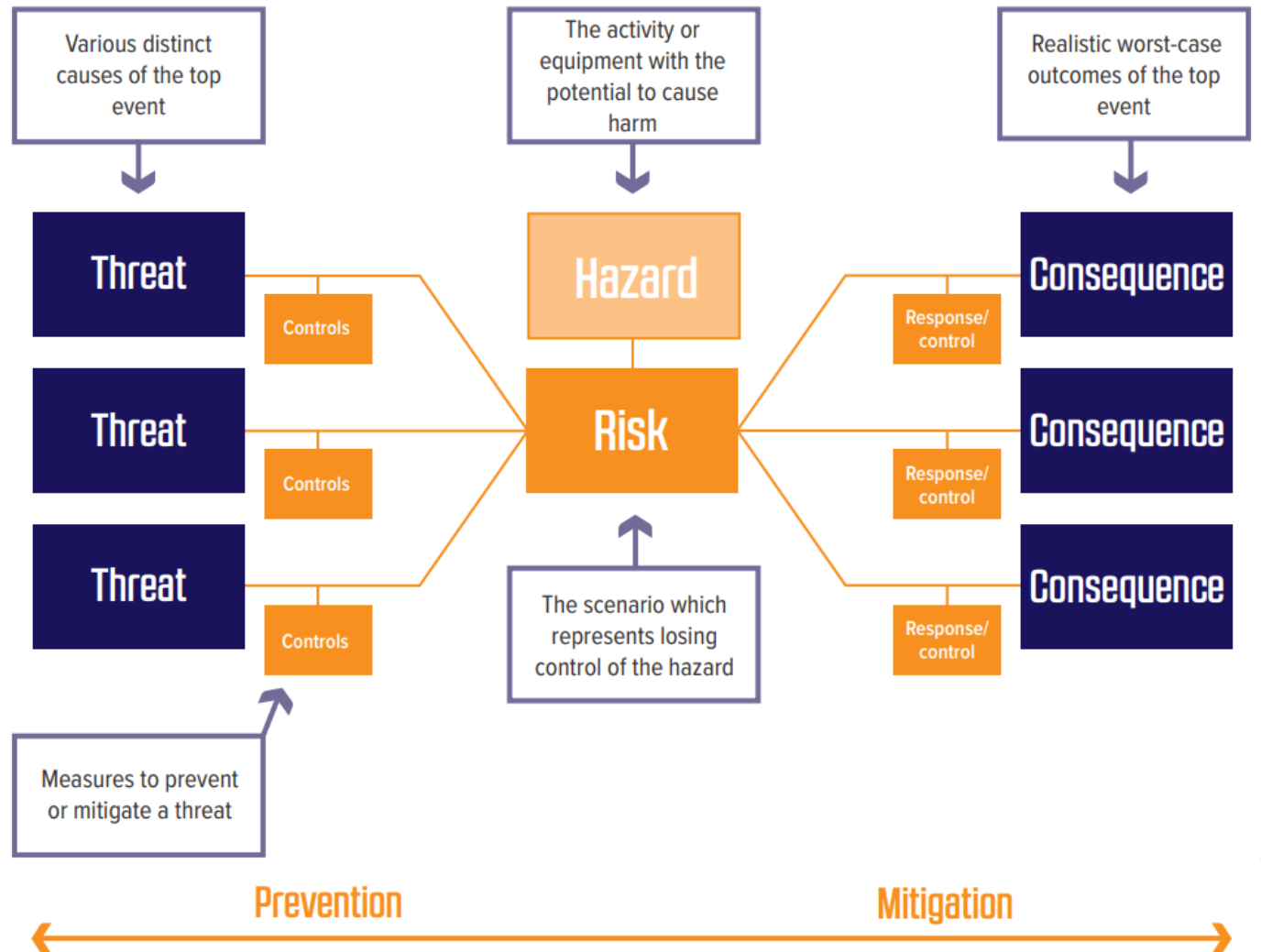
- Severity:** Severity of impact of failure event. It is scored on a scale of 1 to 10. A high score is assigned to high-impact events while a low score is assigned to low-impact events.
- Occurrence:** Frequency of occurrence of failure event. It is scored on a scale of 1 to 10. A high score is assigned to frequently occurring events while events with low occurrence are assigned a low score.
- Detection:** Ability of process control to detect the occurrence of failure events. It is scored on a scale of 1 to 10. A failure event that can be easily detected by the process control is assigned a low score while a high score is assigned to an inconspicuous event.
- Risk priority number:** The overall risk score of an event. It is calculated by multiplying the scores for severity, occurrence and detection. An event with a high RPN demands immediate attention while events with lower RPNs are less risky.

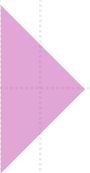
Risk Analysis M

- Bow Tie Analysis

Canterbury Flood - Hazard

BOW TIE ANALYSIS





Making decisions

Risk Matrix

RISK ASSESSMENT MATRIX

		CONSIDER THE LIKELIHOOD OF A HAZARDOUS EVENT OCCURRING				
RISK ASSESSMENT MATRIX		Very unlikely to happen	Unlikely to happen	Possibly could happen	Likely to happen	Very likely to happen
	Catastrophic (e.g fatal)	Moderate	Moderate	High	Critical	Critical
	Major (e.g Permanent Disability)	Low	Moderate	Moderate	High	Critical
	Moderate (e.g Hospitalisation/Short or Long Term Disability)	Low	Moderate	Moderate	Moderate	High
	Minor (e.g First Aid)	Very Low	Low	Moderate	Moderate	Moderate
	Superficial (e.g No Treatment Required)	Very Low	Very Low	Low	Low	Moderate